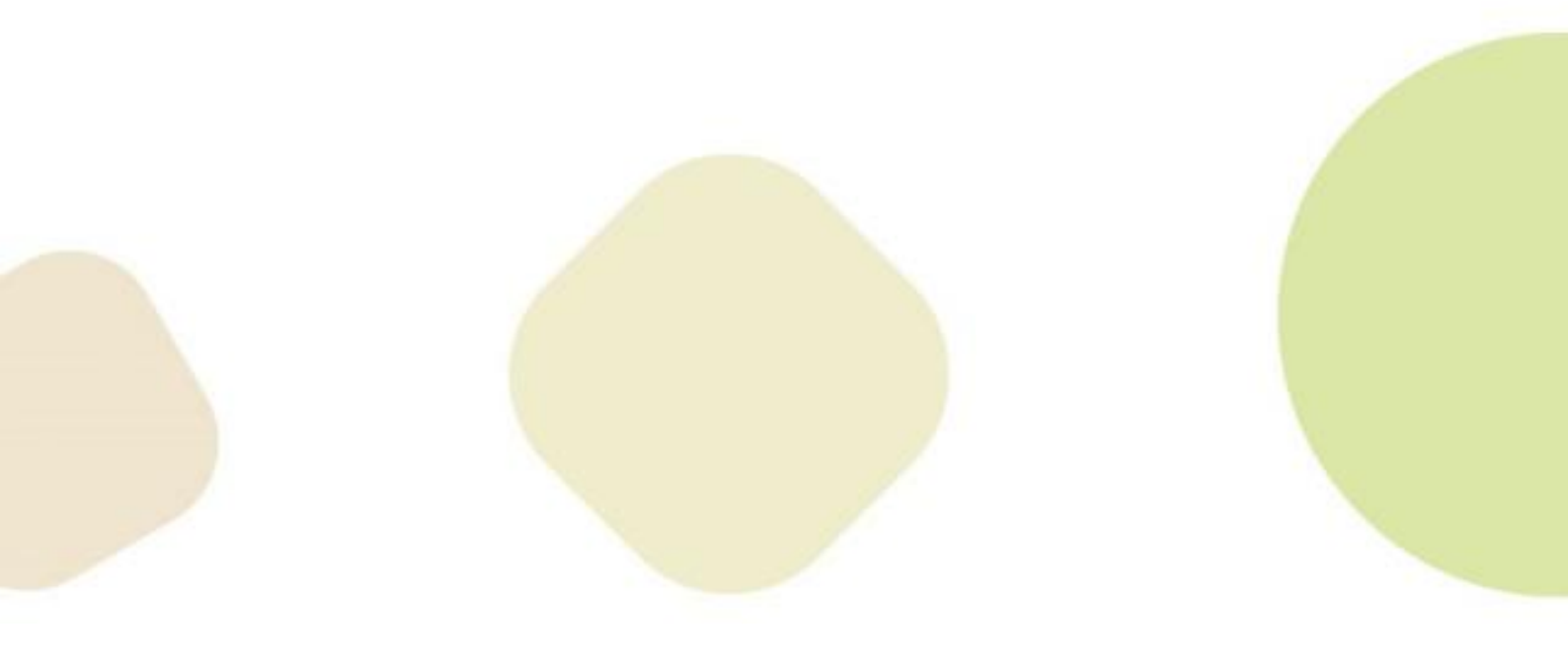




SUSTRACK

D4.1 - REPORT PRESENTING SCENARIOS AND RELATED POLICY PACKAGES



Work package	WP4 Creation, quantification and interpretation of scenarios for the sustainable transition
Task	4.1 Development of policy portfolio scenarios
Due date	31.12.2023
Submission date	29.12.2023
Deliverable lead	Sören Richter
Version	V6.0
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Document Revision History

Version	Date	Description of change	List of contributor(s)
1.0	13.12.2023	Draft version distributed for WP partners' input	DBFZ;KE
2.0	15.12.2023	Draft version distributed for partner's input	UNIT;UGENT;BAM
3.0	21.12.2023	Review's input	Sören Richter (DBFZ)
4.0	22.12.2023	Final version	DBFZ
5.0	30.07.2024	Reopen for Revision	DBFZ, KE
6.0	26.08.2024	Final revised version	DBFZ

Abstract

The transition from linear, fossil-based systems to circular and bio-based systems represents an opportunity and an appropriate pathway to achieve several SDGs. Indeed, circular bio-based systems offer an effective way to reconcile long-term sustainable growth with environmental protection through the prudent use of renewable resources for industrial purposes. This necessary transition is a complex process that requires not only innovative supply-side technologies, but also societal transformations based on a multi-actor process. The circular bioeconomy meta-sector may be a good candidate for proposing a new economic model, requiring transformative policies, targeted innovation, access to finance, risk-taking capacity, and new and sustainable business models and markets. In this respect, a critical assessment of the environmental, social, and economic impacts of the current linear fossil-based economy, as compared to the potential for improvement associated with circular bio-based systems, is needed to inform the identification of policy priorities. Against this background, SUSTRACK is a three-year project that aims to support policy makers in their efforts to develop sustainable pathways to replace fossil and carbon-intensive systems with sustainable circular and bio-based systems (at EU and regional level), thus contributing to the objectives of the European Green Deal. This will be done by: identifying the environmental, economic, and social limits of a linear carbon-intensive and fossil-based economy; improving existing assessment methodologies; assessing the environmental, social and economic impacts of the EU's current linear fossil-based economy; comparing different transition scenarios focusing on the most carbon intensive sectors; identifying priorities according to the scenarios analysed in the project and developing guidelines and policy recommendations.

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EXECUTIVE SUMMARY

The aim of this report (D4.1) is twofold: (i) to identify and describe policy goals that are essential for the transformation to a circular biobased economy (CBBE), and (ii) to develop policy portfolio scenarios for the four focus sectors of the project, namely construction, chemicals, plastics and textiles. This work on policy goals and scenarios supports the development of the Green Economy Model (GEM) used in WP4, and the identification of policy priorities in WP5. The scenarios formulated and presented in this report also support more detailed research to be carried out for the case studies to be analysed in each sector, all along the duration of the project.

In its first section, this report provides an overview of the method (i.e. different steps implemented for data collection and scenario development). Concerning data gathering, screening criteria have been elaborated for the adequate identification of policy goals impacting the evolution of CBBE. Further, a template was prepared for data collection, that supports the implementation of the scenarios in the GEM model (i.e. model inputs) as well as the creation of a portfolio of policy goals. Further, the report provides an overview of the collected and selected policy objectives included in the scenarios, in the form of short fact sheets. Finally, the scenarios are described at the EU level, including sector-specific policy goals as well as more general cross-sectoral and macroeconomic goals. For each focus sector of chemicals, construction, plastic and textile, two scenarios have been developed. The BioTransition scenarios consider existing, *status quo* policy objectives. The BioRevolution scenarios instead integrate new and more ambitious goals, aiming at a rapid transformation towards a fully CBBE.

In summary, the report first presents the methodology adopted for the identification of a portfolio of policy goals and the development of the scenarios (Section 2). The results section provides an overview of the selected policy goals. Based on the screening of 49 policy documents, expert consultations and verification via stakeholder workshops, 73 policy goals were identified and selected (Section 3). Finally, building on these goals, one BioTransition and one BioRevolution policy portfolio scenario at the EU level and applied to each sector (chemicals, construction, plastics and textiles) are presented (Section 4).

TABLE OF CONTENTS

1	Introduction	11
1.1	Objective of the report.....	12
1.2	Structure of the report	12
2	Methods	13
2.1	Scenario scope	13
2.2	Selection of focus countries	13
2.3	Elaboration of screening criteria.....	13
2.4	Template development for policy goals.....	14
2.5	Scenario formulation.....	14
3	Results.....	16
3.1	Scenario scope	16
3.2	Screening criteria	17
3.3	Policy goal data gathering	18
4	BioTransition and Bioevolution scenario	20
4.1	Overarching scenario storyline	20
4.1.1	BioTransition: Transition to a biobased economy.....	20
4.1.2	BioRevolution: Revolution to a biobased economy	20
4.2	Overarching portfolio scenarios based on EU goals.....	21
4.2.1	BioTransition: Transition to a biobased economy.....	21
4.2.2	BioRevolution: Revolution to a biobased economy	22
4.3	Chemical sector	23
4.3.1	BioTransitionCHEM: Transition to a biobased economy	23
4.3.2	BioRevolutionCHEM: Revolution to a biobased economy.....	24
4.4	Construction sector	25
4.4.1	BioTransitionCON: Transition to a biobased economy.....	25
4.4.2	BioRevolutionCON: Revolution to a biobased economy	26

4.5 Plastic sector.....27

4.5.1 BioTransitionPLA: Transition to a biobased economy27

4.5.2 BioRevolutionPLA: Revolution to a biobased economy28

4.6 Textile sector29

4.6.1 BioTransitionTEX: Transition to a biobased economy29

4.6.2 BioRevolutionTEX: Revolution to a biobased economy30

5 Conclusion31

References32

Annex39

LIST OF FIGURES

Figure 1: Methodological steps for the elaboration of the scenario formulation	13
Figure 2: Screening criteria for policies	17
Figure 3: Portfolio of identified policy goals (a) in relation to the reference year; (b) in relation to their geographical characteristic; (c) in their qualitative or quantitative characteristic; (d) with impact on the specific part of the value chain	18
Figure 4: Portfolio of identified policy goals integrated in the scenarios (a) in relation to the reference year; (b) in relation to their geographical characteristic; (c) in their qualitative or quantitative characteristic; (d) having impact on the specific part of the value chain	19
Figure 5: Screening criteria that correspond with identified policy goals of the overarching policy goals on EU level	21
Figure 6: Screening criteria that correspond to identified policy goals of the chemical sector ...	23
Figure 7: Screening criteria that correspond with identified policy goals of the construction sector	25
Figure 8: Screening criteria that correspond with identified policy goals of the plastic sector ...	27
Figure 9: Screening criteria that correspond with identified policy goals of the textile sector ...	29
Figure A2 1: Input collected for the plastic sector in the INSPIRE conference.....	40
Figure A2 2: Input collected for the chemicals sector in the INSPIRE conference.....	41
Figure A2 3: Input collected for the textile sector in the INSPIRE conference.....	42
Figure A2 4: Input collected for the construction sector in the INSPIRE conference.....	43

LIST OF TABLES

Table 1: Template structure for policy goals collection	14
Table 2: Geographical coverage of the two policy portfolio scenarios.....	16
Table A1 1: Selection criteria for countries and interpretation (Economic)	39
Table A1 2: Selection criteria for countries and interpretation (Emissions).....	39
Table A4 3: Overview of portfolio of selected policy goals for the BioTransition scenario of the overarching EU level addressing CBBE aspects.....	91
Table A4 4: Overview of portfolio of selected and more ambitious policy goals for the BioRevolution scenario of the overarching EU level addressing CBBE aspects.....	92
Table A4 5: Overview of portfolio of selected policy goals for the BioTransition scenario for the chemical sector addressing CBBE aspects	93
Table A4 6: Overview of portfolio of selected and more ambitious policy goals for the BioRevolution scenario for the chemical sector addressing CBBE aspects	94
Table A4 7: Overview of portfolio of selected policy goals for the BioTransition scenario for the construction sector addressing CBBE aspects	96
Table A4 8: Overview of portfolio of selected and more ambitious policy goals for the BioRevolution scenario for the chemical sector addressing CBBE aspects	97
Table A4 9: Overview of portfolio of selected policy goals for the BioTransition scenario for the plastic sector addressing CBBE aspects	99
Table A4 10: Overview of portfolio of selected and more ambitious policy goals for the BioRevolution scenario for the plastic sector addressing CBBE aspects.....	101
Table A4 11: Overview of portfolio of selected policy goals for the BioTransition scenario for the textile sector addressing CBBE aspects	104
Table A4 12: Overview of portfolio of selected and more ambitious policy goals for the BioRevolution scenario for the textile sector addressing CBBE aspects.....	106

List of abbreviations	
Abbreviation/Acronym	Description
CBBE	Circular biobased economy
CCU	Carbon capture and use
CO ₂	Carbon dioxide
ETS	Emission trading system
EU	European Union
FAQ	Frequently asked questions
GEM	Green Economy Model
GHG	Greenhouse gas emission
LULUCF	Land use, land use change and forestry
WP	Work Package

1 INTRODUCTION

The bioeconomy is seen as a central approach for reaching the EU's ambition to become climate neutral by 2050 and to support sustainable development (1). After 12 years of strategy development at the EU level, it has become evident that a sustainable bioeconomy will be an enabler for a cross-sectoral transformation that includes all environment, economy and society (2). While the cross-sectoral and multidimensional nature of the bioeconomy is intrinsic, it is already evident that the concept must be tailored to specific country contexts, considering both the availability of raw materials and the efficiency in the use of resources. As a result, there cannot be a single bioeconomy, but rather many, each adapted to different regions and sectors across countries and regions (2).

Although the bioeconomy definition and its implementation may vary significantly across contexts, it is clear that the bioeconomy could support the adoption of sustainable production systems, while also contributing to the mitigation of climate change impacts.

Delving deeper into the definition, while the broader concept of the bioeconomy is still under discussion (3, 4), it is clear that it has to be circular, as defined in the circular bio-based economy (CBBE) concept (5, 6). According to the EU bioeconomy strategy: "(...) To be successful, the European bioeconomy needs to have sustainability and circularity at its heart. (...) " (1). The circular economy is characterised by recycling and reduction objectives in contrast to the linear fossil-based economy (6) and aims to develop an economy "where the value of products, materials, and resources is maintained in the economy for as long as possible, and the generation of waste is minimised (...) " (7). This stresses two central drivers of the transformation process, (i) the fact that bio-based raw material supply is limited, as well as (ii) the decoupling of economic growth from the use of virgin resources (material efficiency) (6, 8, 9).

As a starting point for the evaluation, a sectoral perspective is adopted. This is useful to develop effective governance structures for a CBBE, using a bottom-up approach. Effective governance is needed, as an enabling condition, to scale up biobased concepts, processes as well as products from niche application to sectoral best practice. In this task and report, policies are first contextualized using a macro-level perspective (10), and then sector-level impacts are assessed via the use of a bio-based lens. This information will be picked up by WP4 and WP5 of the project, which focus on estimating the socio-economic and environmental outcomes of policy implementation, using the Green Economy Model (GEM), and on formulating policy packages.

1.1 Objective of the report

This report summarizes the work completed in T4.1, focusing on developing policy portfolio scenarios for CBBE in the EU. The objective of this report is to create and present policy portfolio scenarios¹, including specific sectoral goals by country, to guide the development of GEM. GEM will simulate and assess how well national and regional policy goals align with EU-level transition outcomes.

The report presents and discusses the findings of the scenario development exercise, for the selected case study sectors and countries (related to the work of WP3 and WP5), as well as at the EU level. These scenarios contextualize the case studies and also support the evaluation of existing and new policy goals within the CBBE framework, and generate a portfolio database of key policy goals.

1.2 Structure of the report

Besides this introductory section, the report is structured as follows: Chapter 2 illustrates the methodological approach used for scenario development; Chapter 3 presents the results of the literature review focused on existing policies, for the sectors and countries considered; Chapter 4 illustrates the scenarios and their characteristics; Chapter 5 closes the report by summarising the main findings of T4.1, and outlines the next steps within the SUSTRACK project.

¹ Scenarios are used as a method to describe possible future pathways, resulting from the use of different underlying assumptions. Qualitative scenarios are developed in this report, presented via the use of narratives, or storylines, of alternative possible futures. These scenarios should not be interpreted as predictions or forecasts, nor as an accurate picture of the future. These are systemic, yet partial assessments, and cannot include all the components of a system that are found in reality (11, 13). In other words, scenarios reflect simplifications of reality and its complexity.

2 METHODS

The generation of scenarios followed five key steps, as illustrated in the following overview (Figure 1).

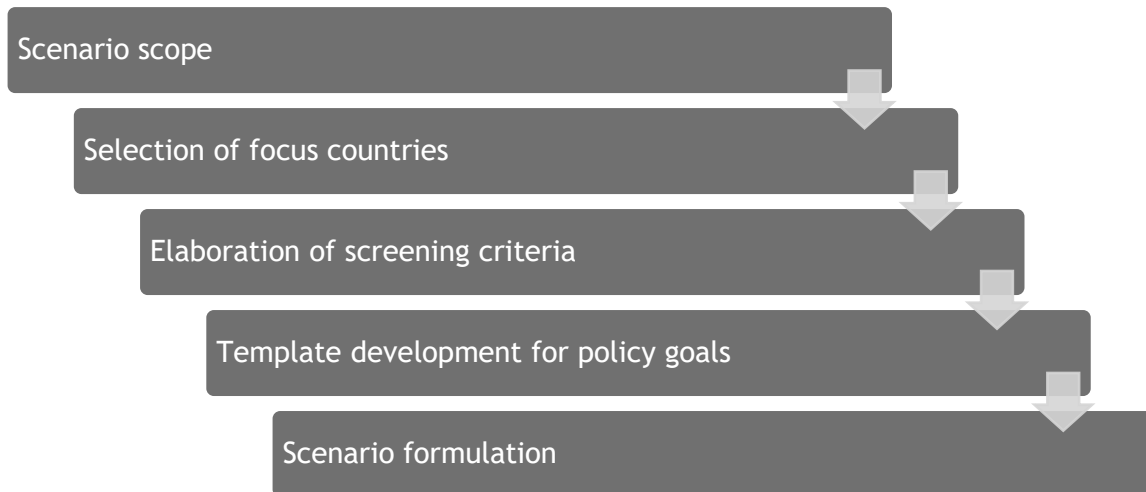


Figure 1: Methodological steps for the elaboration of the scenario formulation

2.1 Scenario scope

The first step in scenario development consists of describing the general scenario scope, along with the quality criteria of plausibility, consistency, comprehensibility and traceability, distinctness and transparency (11, 12). This description of the context contains information about the objectives, project goals, scenario typologies as well as scenario components (13). Specifically, policy goals are identified that support the transformation of selected sectors from a linear to a circular bio-based model. The selected sectors are chemicals, construction, plastics and textiles. The policy portfolio scenarios are based on the current *status quo* of policy at the EU, national and sectoral levels. Concerning future ambition, both existing strategy and policy documents as well as stakeholder feedback, and targets provided were considered.

2.2 Selection of focus countries

The focus countries have been selected based on their potential impact on a transformation to CBBE, applying the following three criteria:

1. Importance of the sector (i.e. contribution of sectors to the national economy and selection of the case studies in previous WPs)
2. Total size of the economy
3. GHG emissions by sector at EU level

2.3 Elaboration of screening criteria

In order to select the appropriate policy documents, screening criteria were established based on the circular economy strategy and the circular economy action plan of the EU (7, 14). Applying

these criteria, the policies for chemicals, construction, plastics and textiles were identified, reviewed and analysed.

For the collection of macro-level goals, the policy field is differentiated into policy objectives, goals and instruments (15). Policy objectives represent desired future conditions or states to be achieved and are often long-term (e.g. 30 % target for the use of recycled plastic in beverage bottles by 2030 (16)). Policy goals, instead, are set for the medium term. Policy instruments are defined as the measures (or intervention options) implemented to realize policy goals and objectives. These are assessed in more detail in WP5, where pathways of change are identified, policy priorities reviewed, and enabling policy instruments proposed.

2.4 Template development for policy goals

The collected policy goals (both of qualitative and quantitative nature) were analysed using specific parameters, identified in an iterative process with the modelling task (T4.2) (see Table 1). The resulting parameters have been included in a template document, used to collect and organize relevant information.

Data was gathered through an extensive literature review, including national and sectoral strategies and directives, existing regulations for CBBE, as well as national energy and climate plans. This desk research and selection of policy goals were further enhanced by expert consultations within the project consortium, as well as with external stakeholders at the INSPIRE conference (see Annex 7.1 for MIRO Boards of the Workshop). During the workshop, stakeholders discussed selected policy goals in group sessions (dedicated conversations took place for each sector). Based on the feedback received, the selection of policy goals was reviewed and improved, and new goals were integrated into the scenarios.

Table 1: Template structure for policy goals collection

Name
Goal name and description
Nature (Quantitative or qualitative)
Mechanism (through which the goal should be realised)
Goal implementation year (2030, 2050, other)
Expected impacts
Geographic level (EU, country of the case studies)
Sector covered (Textile, Construction, Plastic, Chemicals)
Influenced value chain part (production, consumption or end of life)
References

2.5 Scenario formulation

The aim of the scenario formulation task is to generate both a baseline scenario and a more ambitious transformative scenario. Given the systemic nature of the CBBE, a portfolio of policy goals was selected for both scenarios. Further, the selection of policies was informed by a literature review and consultations with experts from the project consortium and at the international INSPIRE conference.

The baseline scenario illustrates the *status quo* of policies identified for CBBE at both EU and country levels. In other words, it considers stated targets and policies already discussed, approved

and either being implemented or soon to be implemented. Additionally, the implementation years of each target are presented, both to assess the complementarity of each intervention option in the context of the broader CBBE, as well as to support the incorporation of quantitative policy targets in the GEM model and related quantitative analysis.

The transformative scenario, a more ambitious case, underlines a faster, more holistic and better-integrated transition towards the CBBE. A comparative table was generated to support the comparison of the two scenarios, listing all policy goals and their implementation years (see Annex A3). Colour coding and additional text for the new and more ambitious policies highlight the differences between the scenarios.

3 RESULTS

3.1 Scenario scope

The two scenarios developed were named BioTransition (baseline scenario) and BioRevolution (ambitious scenario). Based on the country's assessment and selection process (see Annex Table A1 1 and A1 2), Germany, Italy and the Netherlands were selected for detailed analysis in the policy portfolio scenarios.

The scenarios were formulated to cover all sectors, in the EU and for Germany, Italy and the Netherlands, as presented in Table 2. For the BioRevolution scenario cross-country more ambitious goals have been integrated, while for the BioTransition scenario the focus was on the country specific goals. The timeframe extends to 2050 for the chemical and construction sectors and to 2040 for the plastic and textile sectors, as a result of data availability.

Table 2: Geographical coverage of the two policy portfolio scenarios

Sector	BioTransition	BioRevolution
Cross-sectoral	EU	EU
Chemical sector	Netherlands	Netherlands
Construction sector	Germany	Germany, Netherlands, Italy
Plastic sector	Germany	Germany, Netherlands,
Textile sector	Italy	Italy, Netherlands

3.2 Screening criteria

For selecting and identifying the policy goals underlying the scenarios, screening criteria were defined, considering the definition of CBBE used in this project, and building on the EU Circular Action Plan (see Figure 2).



Figure 2: Screening criteria for policies

The criteria are described as follows:

- Broaden material use, of biogenic materials, residues and wastes: Increase the integration of biogenic raw materials, such as primary biomass, sustainable biowastes, or other currently classified waste materials into material flows
- Expand circularity instead of linearity: Every product should be used as long as possible within the same or other value chains and as a raw material for other processes
- Scale down the amount of waste and disposal: Within a circular economy no waste or disposal exists and every residual product should be a raw material for other processes
- Cut down GHG emissions: Lower GHG emissions by decreasing the use of primary raw materials and increasing the integration of renewable raw materials
- Lower biodiversity and land use impacts: Using fewer primary raw materials and enhancing circularity leads to reduced impacts on biodiversity and land use
- Raise efficiency of material usage: A circular economy focuses on closing material loops and reducing the need for new raw materials for products

3.3 Policy goal data gathering

Based on the characterization of the scenarios and the screening criteria, the following overview presents the metrics identified and analysed in two different selection phases.

Figure 3 (a-d) illustrates the results of the first selection of policy goals along important metrics taken from the categories of the template (see chapter 2.4).

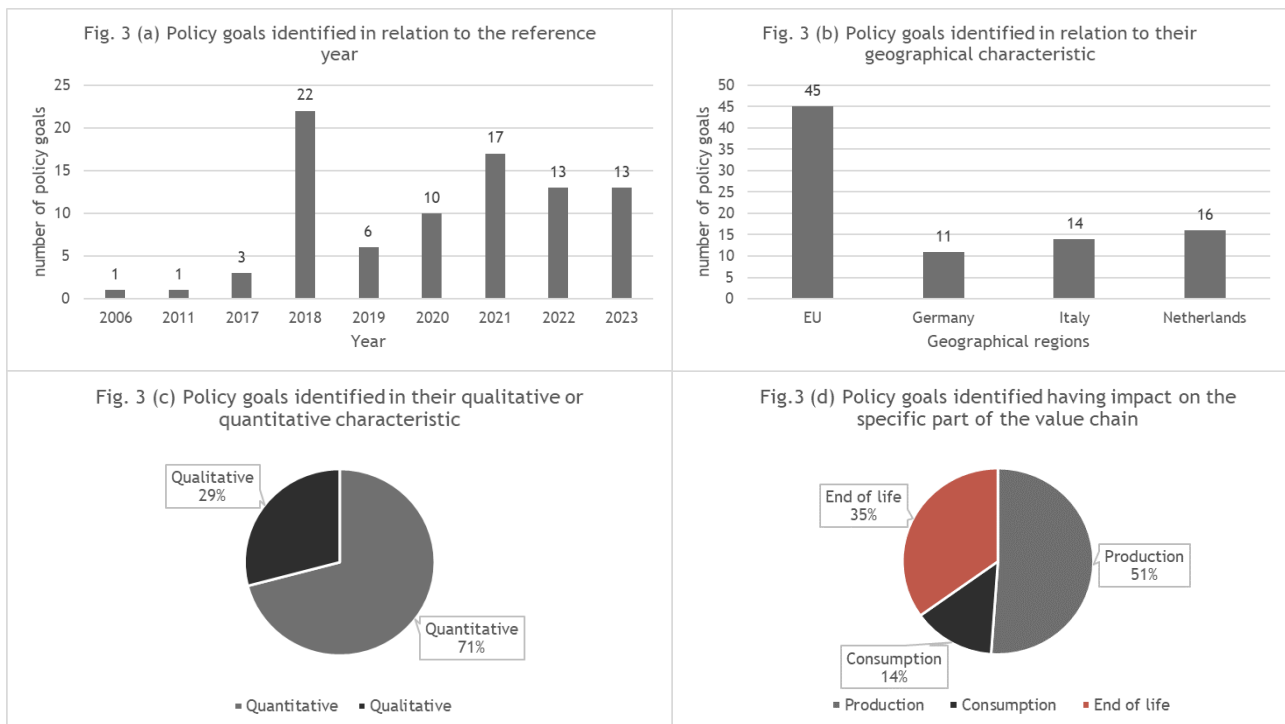


Figure 3: Portfolio of identified policy goals (a) in relation to the reference year; (b) in relation to their geographical characteristic; (c) in their qualitative or quantitative characteristic; (d) with impact on the specific part of the value chain

In this selection process, 86 policy goals were identified, 81 of which are being implemented or under discussion in the period 2018 and 2023. Aligned with the project's focus, most policy goals were identified at the EU level. Additional efforts were made to collect quantitative goals, in addition to qualitative indications of ambition, to support the modelling task in WP4.

We observed that most goals impact all three main components of the value chain: primarily on production, followed by end-of-life considerations, with only a small number of policies and targets focusing on consumption.

Figure 4 (a-d) illustrates the important metrics for the policy goals selected and integrated into the scenarios.

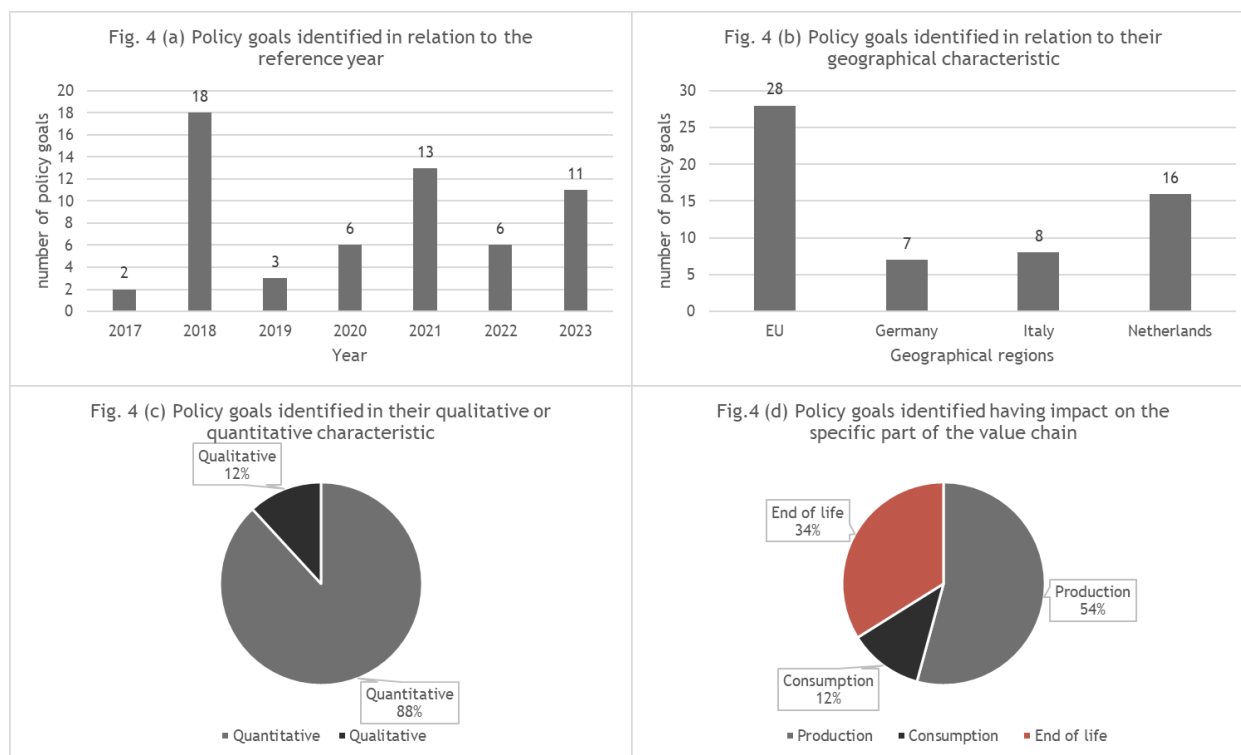


Figure 4: Portfolio of identified policy goals integrated in the scenarios (a) in relation to the reference year; (b) in relation to their geographical characteristic; (c) in their qualitative or quantitative characteristic; (d) having impact on the specific part of the value chain

Through the iterative process of identifying and formulating specific policy goals, 27 goals were eliminated. As a result, 59 policy goals were retained. The majority of policy goals were identified at the EU level, while second most in the Netherlands followed by Italy and Germany. The majority of goals carry a quantitative target. Also, in this case, we find that most policies focus on production, followed by end of life, and finally consumption.

Additionally, certain goals were added via the expert consultations during the INSPIRE Conference. Specifically, 14 additional policy goals were included in the two scenarios, which led to a final total of 73 policy goals analysed and used to elaborate the storyline and the portfolio scenarios of policy goals. A more detailed overview of each collected policy goal is given in Annex A3 in the form of short factsheets, along with the categories of the template developed.

4 BIOTRANSITION AND BIOREVOLUTION SCENARIO

This chapter provides insight into the two scenarios formulated, informed by the policy goals identified and stakeholder input. An initial general overview is followed by a more detailed presentation of the scenarios, considering specific sectoral dynamics across countries (see Chapters 4.1 - 4.3).

Each chapter begins with an overview of the criteria, and how these are addressed by the policy goals in the *status quo* scenario, followed by the storyline. The table listing specific policy goals and their timeframe is presented in Annex A4.

4.1 Overarching scenario storyline

4.1.1 BioTransition: Transition to a biobased economy



The BioTransition scenario focuses on current policy objectives. Transitions are multi-dimensional processes involving significant structural changes across technological, material, economic, or political elements (17) and their systemic interrelations (18). In the BioTransition scenario, the structural change is based on policy goals that promote aspects of CBBE and its overall system integration. Specifically, it includes quantified goals aimed at altering material flows, waste streams, and end-of-life options, with the aim to increase homogeneous material streams rather than heterogeneous ones. While this scenario establishes the foundations for a circular economy, it does not include additional measures to accelerate the transition to a CBBE in terms of speed and scope.

4.1.2 BioRevolution: Revolution to a biobased economy



The BioRevolution scenario features new and more ambitious goals aimed at rapidly transforming towards a fully CBBE. The term “Revolution” signifies a faster and more comprehensive transformation, addressing large-scale societal changes across multiple dimensions (18). This scenario incorporates state-of-the-art and forward-looking policies that extend beyond basic waste separation to propose more holistic approaches and drive deeper systemic shifts. It aligns with the new circular economy action plan (14), which focuses on reforming the EU waste legislation (19).

In this storyline, policies address the entire value chain, from resource sourcing to consumption and end-of-life materials management. By adopting more ambitious quotas, setting voluntary measures as mandatory and adjusting time scales for policy goal implementations, the circularity of the material flows is enhanced. Increased policy ambition strengthens political support for transitioning to a CBBE, amplifying the positive climate mitigation impacts of circular material flows.

4.2 Overarching portfolio scenarios based on EU goals

The policy portfolio scenarios applied to the four industrial sectors include two different levels of policy goals and these correspond to 5 out of 6 screening criteria directly (Figure 5). First, the overarching EU policy goals, which are not specifically formulated for a given sector, but are common to all sectors and set the overall boundary conditions. Second, the sector-specific policy goals, leading to unique sector policy portfolio scenarios.



Figure 5: Screening criteria that correspond with identified policy goals of the overarching policy goals on EU level

4.2.1 BioTransition: Transition to a biobased economy

In the overarching storyline of the scenario, three leading areas of change could be identified, in line with the identified policy goals.

The first area is energy efficiency (corresponding with screening criteria “Raise efficiency (of material usage)”). This supports the creation of a modern, resource-efficient and competitive economy (20) oriented towards the goal of no net emissions (21).

The second area focuses on the reduction of waste streams at different levels, but with a focus on municipal waste (e.g. reduction >55% by 2025 and >60% by 2035) (22, 23). Municipal waste includes (see 2.b in (23)) plastics, biowaste, wood, textile, packaging and furniture, in alignment with the sectors considered.

The third area covers the reduction of GHGs (24), with emphasis on specific climate gases such as CO₂ (25) (e.g. 55% by 2030), as well as curbing the environmental impacts (22) generated by material and energy flows (e.g. reduction of final energy consumption 36% by 2030, renewable energy of 42,5%). The aim is to reduce carbon dependency, recycle carbon from waste streams to

replace fossil-based carbon, and upscale carbon removal solutions, such as increasing the use of wood products (25, 26). The aim is to reduce the use of non-renewable resources and strengthen the economic competitiveness of European countries (1), primarily by working within planetary boundaries (e.g. decoupling economic growth from resource use; reducing dependency on non-renewable unsustainable raw materials) (22). Table A4 in Annex 3 provides more detail on the policies and targets considered.

4.2.2 BioRevolution: Revolution to a biobased economy



In the BioRevolution scenario, the push of CBBE is more systemic, establishing specific goals also in the material sector. By addressing the importance of connecting the material flows with the overall transformation into CBBE, it is possible to generate knowledge about the necessary actions that are required to accelerate the transformation to a sustainable system. A central aim of CBBE is to reduce material use and increase efficiency, thereby establishing absolute decoupling of resource use from economic growth (21, 27). This supports also the policy goals that address the issue of total raw material productivity (see BioRevolution Plastics and Construction), enable a more intensive use of secondary raw materials and integration of more sustainable biobased alternatives that are part of natural CO₂ circles (see Construction sector: BioRevolution). A further goal is the integration of renewable raw materials to substitute fossil or non-renewable resources (e.g. increase of wood-based products in the global market to 1,5%). For a detailed overview of included policies sorted by the respective focus year see Annex 4, Table A4.

4.3 Chemical sector

The chemical sector is an upstream producer for several other sectors, for instance construction, textile and plastics and the policy goals correspond to 4 out of 6 screening criteria directly (Figure 6). This characteristic differentiates it from the other downstream sectors. As the sector is not the main producer of final products, end-of-life policy goals such as separation of waste streams during collection are not directly applicable.



Figure 6: Screening criteria that correspond to identified policy goals of the chemical sector

4.3.1 BioTransitionCHEM: Transition to a biobased economy



In the BioTransition scenario, the main goal for the chemical sector is the European Communique's goal of integrating 20 % carbon from non-fossil resources by 2030. This involves reducing fossil carbon and exploring the use of new carbon sources for industrial use. It also includes capturing between 300 Mt and 500 Mt of carbon (25). Another key objective is to ensure toxic-free material flows to support circular material flows and reduce environmental impacts (28). The goals align with the selection criteria "Broaden material use of biogenic materials, residues and wastes" and "Expand circularity instead of linearity". The BioTransitionCHEM scenario emphasizes the substitution of products through biobased alternatives.

Country-specific aspects: For the chemical sector, the focus is on the Netherlands. No specific policies for CBBE in the chemical sector were found. Therefore, the analysis incorporates overarching goals that impact the sector. The latest National Circular Economy Program (29) proposes quantitative goals, such as reducing the raw material footprint (reduction of 50% primary raw material usage) (30). For a detailed overview of the included policies by focus year, see Annex 4, Table A4 5.

4.3.2 BioRevolutionCHEM: Revolution to a biobased economy



In the BioRevolutionCHEM scenario, the main innovation is the increased integration of carbon from non-fossil resources and the early achievement of the zero-pollution goal. Using biomass in a chemical CBBE has a more significant impact on industrial resource use than switching to renewable energy. This is especially evident with ambitious goals for reducing primary resource use. In the BioRevolutionCHEM scenario, the adoption of these practices reduces the dependency on fossil resources and curbs environmental pollution. According to a meta-analysis (31), several studies suggest that a higher biomass use is feasible. Aligned to these results, the scenario increases to 30% of the quota for renewable carbons, based on biomass, in 2030 (average value in the meta-analysis) and to 40% in 2050 (focus on the higher values of the scenario studies). Here, trade-offs across all bioeconomy sectors need to be considered, including potential impacts on the EU's carbon sinks (1) and types of the biomass used. A feedstock change would be triggered for the industry, although potential bottlenecks from increasing use in other sectors (e.g. EU Biodiversity Strategy (32)) should be considered. Chemical pollution significantly impacts environmental quality and biodiversity (32). In the more ambitious scenario, the zero-pollution goal (28) is achieved by 2045. and the goal to reduce substances of concern in recycled materials will be met by 2040. For the detailed overview of included policies see Annex Table A4 6.

4.4 Construction sector

Construction is one of the most material-intensive sectors, but also carries considerable potential to integrate more biobased materials. In particular, the use of wood-based construction parts could have meaningful impacts on the transformation of raw material use in the sector. The policy goals correspond to 5 out of 6 screening criteria directly (Figure 7).



Figure 7: Screening criteria that correspond with identified policy goals of the construction sector

4.4.1 BioTransitionCON: Transition to a biobased economy



In the BioTransitionCON scenario, the market share of wood-based products is relatively low compared to non-metallic raw materials (26). Increasing the use of wood-based products could significantly benefit carbon sequestration by linking the construction sector with carbon farming initiatives (25). This aligns with the criteria of “Broaden material use of biogenic materials, residues and wastes”, and “Cut down GHG emissions”. However, since most policy goals focus on energy-related aspects (e.g. renovation quotas for buildings (20, 33): such as at least 2% of annual energy renovation by 2030; residential and non-residential buildings, decarbonised building stock by 2050) only limited impacts on the material side is assumed in this scenario. The material focus is on separating material streams at end-of-life (“Scale down waste and disposal”) and establishing secondary raw material sources for further use (“Expand circularity instead of linearity”) (e.g. 70% by weight from 2020 prepared for recycling or other material recovery). This approach should also contribute to reducing overall GHG emissions by decreasing primary extraction emissions.

Country-specific aspects: For the construction sector, the focus country is Germany. The goal of supporting innovations within the construction of wood-based buildings (34) and the integration of other types of wood is noted qualitatively. The central policy goal is to increase total raw material productivity by 1,5 % annually, as outlined in the third resource efficiency program (35). This aligns with the criteria of “Raise efficiency of material usage”. The goal of greater resource efficiency and resource substitution embeds the principle of the circular or closed-loop economy

in production processes, similar to the scenario in the chemical sector (35). For a detailed overview of included policies sorted by the respective focus year, see Annex Table A4 7.

4.4.2 BioRevolutionCON: Revolution to a biobased economy



In the BioRevolutionCON scenario, the market share of wood-based products increases, aligned with the assumptions of a future worldwide bioeconomy market expansion (26) from 1 % to 1,5 %. The rise of the quota is possible to generate from EU resources, although with possibly high impacts on the competing uses of roundwood (1) and cross-cutting sectors.

An additional component in the BioRevolutionCON scenario is the shift of the EU Taxonomy goals from being voluntary to mandatory. This assumes that for a shift in the construction sector from a carbon source into a sink, the EU Taxonomy rules are necessary to be implemented as a directive starting in 2030. Mandatory product passports support the monitoring of the implementation of these goals. Additionally, the sustainable carbon cycle goal (25) supports the deployment of bio-based materials in the sector. Included in the regulations is also the carbon storage conservation in the vacancy management of buildings, aiming at prolonged usage for old buildings.

Country-specific aspects: The focus country for the construction sector is Germany. Named policy goal incentives are the promotion of innovation in the construction of wood-based buildings (34) and the integration of other types of wood. In the BioRevolutionCON scenario, the total raw material productivity is raised from 1,5 % to about 2 % (35), because the goal of 1,5 % was not reached in the time period from 2010 - 2018, and more ambitious goals are necessary to reach the aim for 2030. Possible measures for this could be higher recycling quotas and increased support of lightweight constructions (36). This is similar to the Netherlands' goals of (i) reducing the usage of abiotic materials until 2030 and (ii) integrating as many sustainable renewable raw materials in products. For a detailed overview of the included policies by focus year, see Annex Table A4 8.

4.5 Plastic sector

The plastics sector has the most specific policy goals based on EU regulation and at the national level and these correspond to 5 out of 6 screening criteria directly (Figure 8). It is a key sector for CBBE transformation due to its high dependency on fossil fuel feedstocks. Therefore, specific concepts in this sector have a high impact on the development of a CBBE.



Figure 8: Screening criteria that correspond with identified policy goals of the plastic sector

4.5.1 BioTransitionPLA: Transition to a biobased economy



The BioTransitionPLA scenario focuses on the reduction of waste streams, by increasing recycling (37), banning products (38), as well as integrating recycled content in the material flows (39). Further regulations address the negative influence on human health and ecosystems, by regulating specific ingredients of the products (39) (e.g. lower the sum of concentration levels of lead, cadmium, mercury and hexavalent chromium).

Although voluntary, the EU Taxonomy is seen as a signal of a shift in mindset in the sector. It sets goals for recycled content in products (e.g. at least 65 % of the packaging product by weight consists of recycled post-consumer material by 2028), the introduction of biobased feedstocks (e.g. use of bio-waste feedstock: at least 65 % of the packaging product by weight consists of sustainable bio-waste feedstock by 2028) as well as the overall decrease of per capita packaging waste generation ((a) 5 % by 2030; (b) 10 % by 2035; (c) 15 % by 2040) (40). Nevertheless, the uptake of this Taxonomy is limited in terms of the number of companies participating, which is quantified between 1-5 % (41). The portfolio of policies in this sector already impacts several parts of the value chain from production, over consumption to end-of-life aspects.

Country-specific aspects: The focus country for the plastic sector is Germany. The central policy goal is to increase total raw material productivity by 1,5 % annually, as outlined in the third resource efficiency program (35). Additionally, recycling quotas are established to accomplish the legislative quotas (3,8 % by 2025; 8,8 % by 2030), set in place by the EU (37). In this sector, the scenario highlights that the EU level has a high impact on the transition of the sector, while on the selected country level, the focus is still on the separation of waste streams and the increase of recycling. For a detailed overview of the included policies by focus year, see Annex Table A4 9.

4.5.2 BioRevolutionPLA: Revolution to a biobased economy



In the BioRevolutionPLA scenario, recycling quotas are specified, more plastic categories are subject to regulation, and mandatory requirements for the production process—especially regarding design—will be introduced. Consequently, certain quotas are increased and bio-based feedstocks are more widely integrated into value chains. The policy goal of aligning the production of plastic products, along with the design for recycling principles, will be established by the year 2025 (39). This will impact recyclability and reduce negative environmental impacts (14). Additionally, the EU Taxonomy policy goal will be mandatory (40, 41) and sets the target for 65 % of packaging products by weight to be made from sustainable bio-waste feedstock.

Country-specific aspects: The focus country for the plastic sector is Germany. The central policy goal is to increase total raw material productivity by 1,5 % annually, as outlined in the third resource efficiency program (35) and is raised from 1,5 % to about 2 % (35), because the goal of 1,5 % was not reached (36) in the time period from 2010 - 2018, and more ambitious recycling quotas are necessary to reach the aim for 2030. Together with the implemented per capita policy goals for the reduction of packaging waste (39), the aim of the EU legislative for recycled plastic should be reached (37). The included policy goals are adapted by the Netherlands' goal for annual reduction in the growth of plastic: 1,5 % annually (42), and address the issue of increasing plastic waste streams (43), and illustrate that both the production level and the consumption level have to be integrated much stronger. Besides the packaging sector, other sectors are also important for a transformation of plastic usage such as the agriculture sector and are also considered in the scenario.

At the country level, the recycling quota for plastics was already at 48,4 % in 2021(43). To accelerate the transformation to a CBBE for the year 2030, an increase of 5 % of the current policy goals is included, leading to the new goal of 60 % in 2030 for plastics recycling and 35 % for wood packaging recycling. An additional measure is the adaption of mechanical and chemical recycling. Aligned to the policy goals of the Netherlands (mechanical recycling to 50 %; chemical recycling to 10 % until 2030 (42)) and evaluations on the EU level (40 % and the chemical recycling to 20 % until 2050 (27)), the scenario integrates quotas for mechanical recycling of 45 % and for chemical recycling of 15 %. By the year 2050, fossil-based production should be phased out as much as possible, with plastic production relying on recycled plastic, biobased materials or Carbon capture and use (CCU) applications. To achieve this goal, it is necessary to consider the limitation concerning the amount of plastics that are actually used and available for recycling, given the reduction assumed in the BioRevolutionPLA scenario. For a detailed overview of the included policies by focus year, see Annex Table A4 10.

4.6 Textile sector

The textile sector is characterized by early action aimed at increasing the share of circular material flows. Since the EU is one of the major consumers of textiles, it is crucial to achieve higher circularity in textile material flows to reduce consumption, which contributes to waste stream problems and negative environmental impacts (48). The policy goals correspond to 3 out of 6 screening criteria directly (Figure 9).



Figure 9: Screening criteria that correspond with identified policy goals of the textile sector

4.6.1 BioTransitionTEX: Transition to a biobased economy

The BioTransitionTEX scenario focuses on the collection and reuse of textiles at the point of disposal (23, 45). The latest strategy of the EU does not set mandatory policy goals but indicates growing ambition for increased use of recycled fibres, absence of hazardous substances and production processes, and ensuring social rights and environmental considerations by the year 2030 (47). Key actions involve implementing comprehensible eco-design requirements, stopping the destruction of unsold and unused fabrics, tackling microplastic pollution (in connection to the Zero Pollution Action Plan (22)), information requirements for the customer, green claims and establishing extended producer responsibility schemes. Addressing these issues requires the monitoring of material flows, starting with the broader implementation of separate textile collection in the waste stream (23).

Country-specific aspects: The focus country for the textile sector is Italy. The central policy goal aims at a 100% recovery in the textile industry by 2050 (45). The Italian National Circular Economy Strategy (46), supports this goal with a focus on the introduction of a tax system favourable to the secondary market. Additional upstream circularity strategies are planned, including eco-design requirements and extending the lifespan of products and parts (47). Given these goals are more

qualitative in nature, in the BioTransitionTEX scenario the aim is to introduce a 100% recovery of textiles by 2026 and reduce the annual energy consumption of the non-emission trading system (ETS) sector (48). For a detailed overview of the included policies by focus year, see Annex Table A4 11.

4.6.2 BioRevolutionTEX: Revolution to a biobased economy



In the BioRevolutionTEX scenario, the goals of the EU Textile strategy are implemented and additional recycling targets are introduced. For clothing, the new policy goals are oriented along the National Circular Economy Strategy of the Netherlands, which integrates specific quantitative goals (29). The adoption of the country-specific goals at the EU level is likely to take more time for the implementation. Thus, the first policy goals will be put in place about 2,5 years later, or by the year 2027. This corresponds with the Italian national goal of 100 % recovery in “Textile Hubs” (45). Other goals are also anticipated in 2030 (e.g. separation of waste streams, recycling quotas), based on the experiences of other sectors (e.g. plastic). Additional measures are addressing directly the consumption side, with a focus on higher use rates of textiles that directly impact climate change, urban land occupation or natural land transformation (49). The goal of fully recycling, reusing, or refurbishing unsold textiles reduces the consumption of new textiles and prevents raw material waste. Along with the obligation to use more recycled clothing, this also supports other goals by 2030.

Besides consumption and end-of-life status, further aspects address the production processes of textiles in the BioRevolution TEX scenarios. In the year 2030, wool production should be qualified and sourced from regenerative agricultural practices. In connection with measures that promote reuse and a sharing economy, natural land transformation and land occupation impacts are addressed. In relation to this full product delivery chain perspective, a carbon tax implemented for all fashion products increases further the reuse of textiles and is a favourable fiscal policy to support a secondary market establishment in the year 2035 (47). For a detailed overview of the included policies by focus year, see Annex Table A4 12.

5 CONCLUSION

This report presents the results of the work carried out in T4.1 of SUSTRACK, which focuses on reviewing existing policy goals, developing portfolio policy scenarios, and identifying goals to accelerate and intensify the transformation to a circular biobased economy.

A total of 49 policy documents were reviewed to identify relevant policy goals. The review emphasized policy goals with both qualitative and quantitative information from a macro perspective, guiding the inclusion of measures in GEM for scenario implementation in T4.2.

The scenario formulation exercise resulted in four main outputs, as presented next:

- First, we identified 86 relevant policy goals at both the EU and country levels for CBBE. 59 of these were integrated into the scenarios. An overall framework with screening criteria (see chapter 2.3) was established to select appropriate policy goals creating an extensive portfolio that informs the modelling task and provides a compendium of policies, particularly quantitative ones.
- Second, the development of policy goals should rely on various inputs, including from sectoral practitioners, literature review and expert knowledge.
- Third, the policy portfolio scenarios offer insights into sectors crucial for the transformation to CBBE:
 - EU Level: Emphasizes the importance of reducing material use while increasing efficiency, aiming to decouple resource consumption from economic growth. CBBE offers a chance to lessen reliance on non-renewable, unsustainable resources, while also promoting the creation of a toxic-free environment.
 - Chemicals Sector: Highlights the goal of increasing the share of non-fossil-based carbon in material flows to support the earlier achievement of the zero-pollution goal, reduce substances of concern in recycled materials, and align with overarching EU objectives.
 - Construction Sector: Demonstrates how a higher share of bio-based products can transform carbon sources into carbon sinks. It also focuses on enhancing raw material productivity and the use of secondary materials to improve material flow circularity, reducing dependence on non-abiotic resources.
 - Plastics Sector: Encourages greater ambition and emphasizes the sector's crucial role in the circular economy. It suggests faster adoption of feedstock changes, setting more specific recycling quotas, and integrating non-fossil-based materials as a model for other sectors.
 - Textiles Sector: Focuses on enhancing resource efficiency by streamlining waste streams and incorporating recycled products and sustainable textiles into value chains to drive sectoral transformation.
- The policy portfolio scenarios highlight the importance of governance structures in facilitating a faster, more holistic transformation of material flows in Europe. The CBBE is crucial for achieving climate goals related to mitigation and adaptation, transforming production and consumption to become more sustainable, focusing on circularity, decoupling resource use from economic growth, and reducing negative environmental impacts.

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Project Consortium



ANNEX

Table A1 1: Selection criteria for countries and interpretation (Economic)

Sectors	Selection criteria for countries and interpretation	Sources
Chemicals	Germany and France are the two largest chemicals producers in Europe valued at €274.7 billion, followed by Italy and the Netherlands. These four countries accounted for 66.2% of EU27 chemical sales in 2021, valued at €393.0 billion. The share rose to 84.3%, or €500.3 billion, when including Spain, Belgium, and Austria. The remaining 20 Member States of the EU accounted for 15.7% of EU27 chemical sales in 2021, with Sweden and Poland being the biggest among them.	(64)
Construction	Annual production value of the construction industry in selected European countries 2020 In billion EUR Top 6: Germany (360,27); France (318,93); Italy (168,2); Spain (140,56); Poland (91,01); Netherlands (85,42)	(65)
Plastics	Plastics converted demand Top 5: Germany, Italy, France, Spain, Poland and for Consumption and Production Top 5: Germany, France, Spain, Netherlands, Belgium	(66, 67)
Textile	The biggest producers in the industry are Italy, France, Germany, Spain and Portugal. Together, they account for about three-quarters of EU production.	(68, 69)

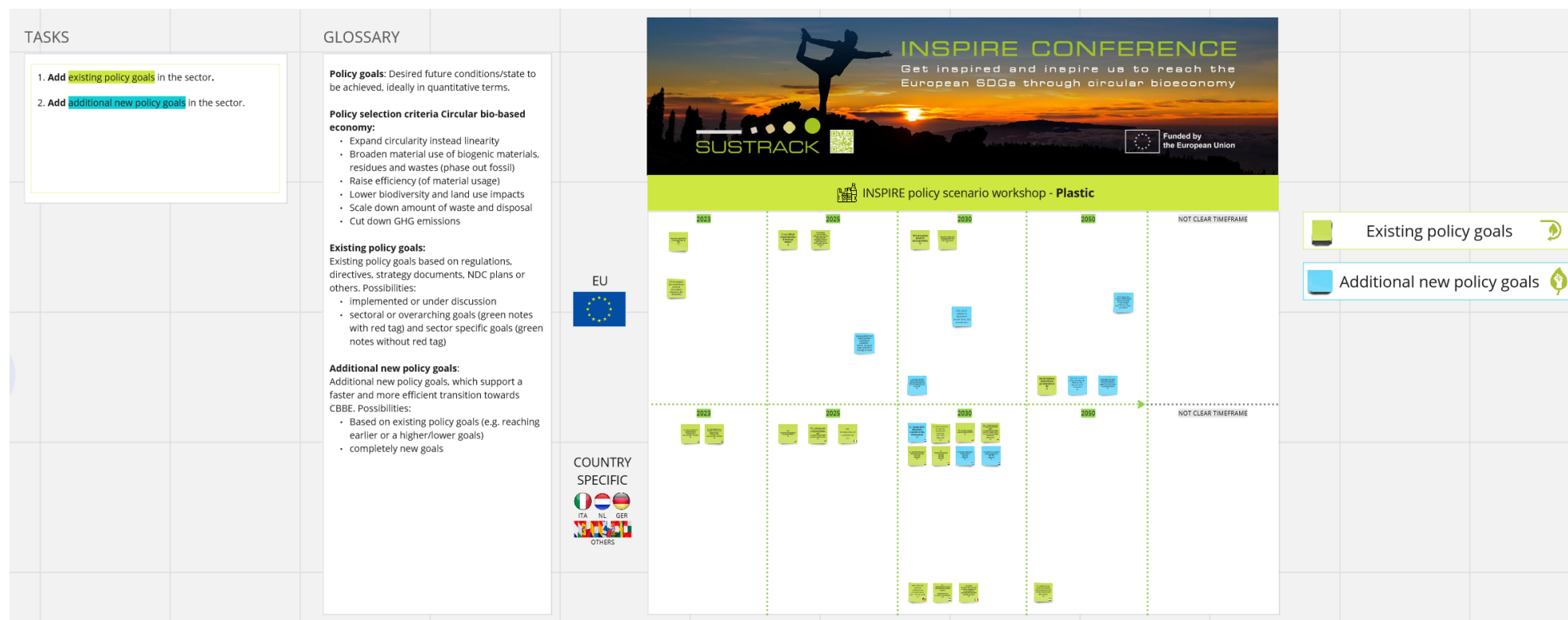
Table A1 2: Selection criteria for countries and interpretation (Emissions)

Description of criteria	GHG-Emission	Sources
Total country GHG emissions in 2019	The top-countries are following: 1. Germany; 2. France; 3. Italy; 4. Poland; 5. Spain; 6. Netherlands; 7. Czech Republic; 8. Belgium	(70)
Chemical sector GHG emissions in 2019 (including plastics)	The top-countries are following: 1. Belgium; 2. France; 3. Netherlands; 4. Germany; 5. Poland	(70)
Manufacturing industries and construction sector GHG emissions in 2019	The top-countries are following: 1. Germany; 2. Italy; 3. France; 4. Spain; 5. Poland; 6. Netherlands	(70)

D4.1. Report presenting scenarios and related policy packages



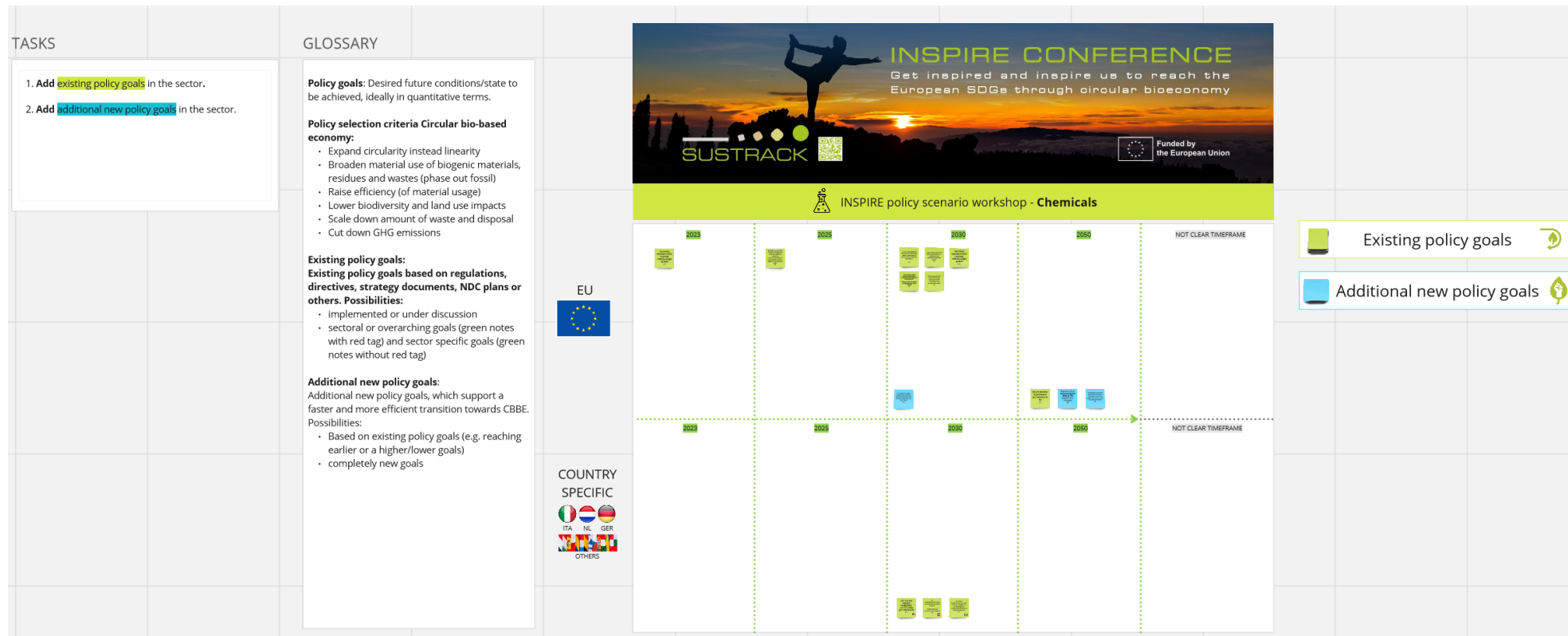
Figure A2 1: Input collected for the plastic sector in the INSPIRE conference



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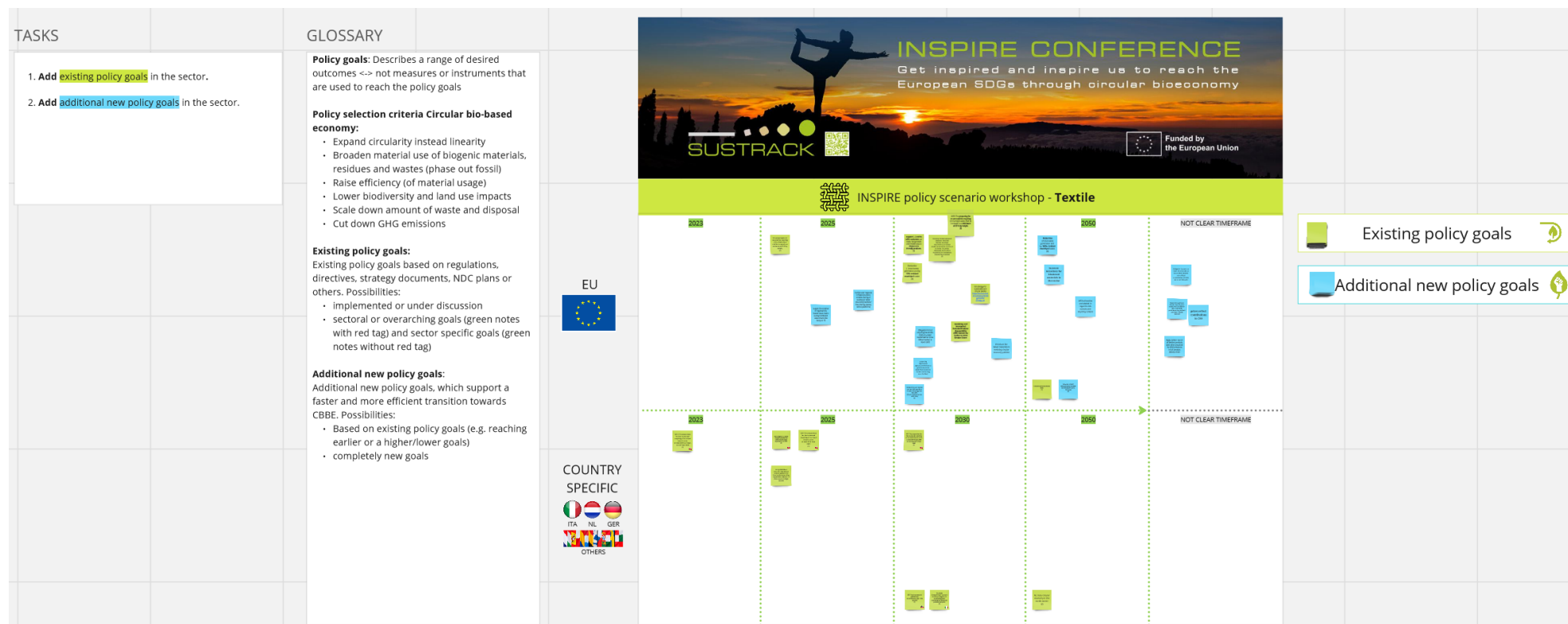


Figure A2 2: Input collected for the chemicals sector in the INSPIRE conference



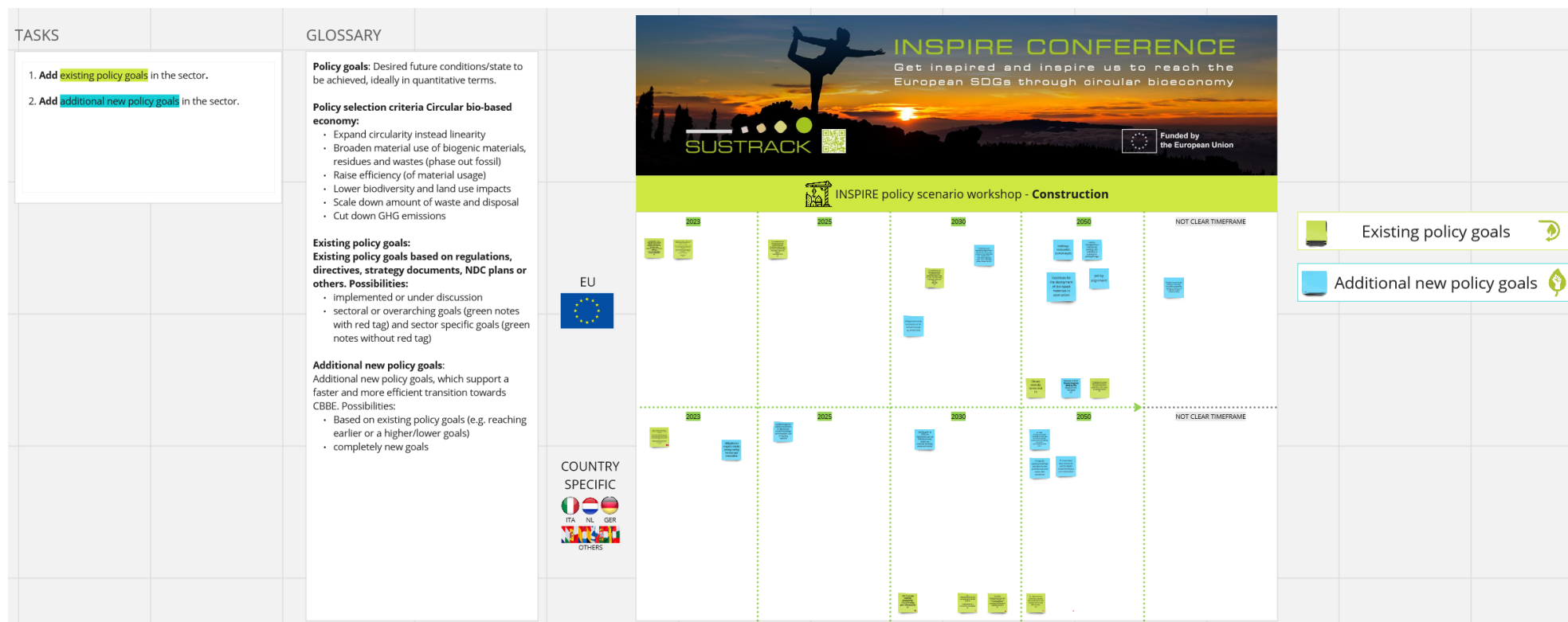
D4.1. Report presenting scenarios and related policy packages

Figure A2 3: Input collected for the textile sector in the INSPIRE conference



D4.1. Report presenting scenarios and related policy packages

Figure A2 4: Input collected for the construction sector in the INSPIRE conference



A3 List of selected and integrated policy goals

A3.1 EU Overarching goals

Policy document name	The European Green Deal (21)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	No net emission of greenhouse gas emissions in EU by 2050 where economic growth is decoupled from resource use.
Mechanism (through which the target should be realised)	1	Quota
Expected Impact	1	(...) aims to transform the EU into a fair and prosperous society, with a modern, resource-efficient and competitive economy where there are no net emissions of greenhouse gases in 2050 and where economic growth is decoupled from resource use. (...) aims to protect, conserve and enhance the EU's natural capital, and protect the health and well-being of citizens from environment-related risks and impacts. At the same time, this transition must be just and inclusive.
Year specified	1	2050
Part of the value chain addressed	1	Production

Policy document name	New EU Forest Strategy for 2030 (26)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	(...) Increase of market share for wood products for the support of reaching climate goals.

Mechanism (through which the target should be realised)	1	The Commission will develop a 2050 roadmap for reducing whole life-cycle carbon emissions in buildings. In the context of the revision of the Construction Products Regulation , the Commission will develop a standard, robust and transparent methodology to quantify the climate benefits of wood construction products and other building materials.
Expected Impact	1	(...) role of wood products is to help turn the construction sector from a source of greenhouse gas emissions into a carbon sink.
Year specified	1	2030
Part of the value chain addressed	1	Production

Policy document name	Proposal for a DIRECTIVE OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL on energy efficiency (recast) (20)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	At least 32.5% improvement in energy efficiency by 2030; Reducing primary (39%) and final (36%) energy consumption by 2030
	2	Annual energy savings obligations by MSs: 2024-2030: >1.5%
Mechanism (through which the target should be realised)	1	Quota
	2	Quota
Expected Impact	1	(...) Energy efficiency is a key area of action, without which the full decarbonisation of the Union´s economy cannot be achieved.
	2	(...) Energy efficiency is a key area of action, without which the full decarbonisation of the Union´s economy cannot be achieved.
Year specified	1	2030
	2	2030

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Part of the value chain addressed	1	Production
	2	Production

Policy document name	The European Climate Law (24)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	Net reduction of GHG emissions by at least 55% (By 2030 vs. 1990 level) (225 million tonnes of CO2 equivalent)
Mechanism (through which the target should be realised)	1	Law
Expected Impact	1	This Regulation establishes a framework for the irreversible and gradual reduction of anthropogenic greenhouse gas emissions by sources and enhancement of removals by sinks regulated in Union law
Year specified	1	2030
Part of the value chain addressed	1	Production

Policy document name	Renewable Energy Directive (27)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	(...) Overall Union renewable energy target to 42,5 % in the energy mix

Mechanism (through which the target should be realised)	1	Binding goal
Expected Impact	1	Reaching intermediate target of a reduction of net greenhouse gas emissions by at least 55 % compared to 1990 levels by 2030
Year specified	1	2030
Part of the value chain addressed	1	Production

Policy document name	Directive 2008/98/EC of the European Parliament and of the Council of 19 November 2008 on waste and repealing certain Directives (23)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	Increasing municipal waste recycling: >55% by weight by 2025 >60% by weight by 2035
	2	Article 8a (4c(ii)) in the case of extended producer responsibility schemes established to attain waste management targets and objectives established under legislative acts of the Union, the producers of products bear at least 80 % of the necessary costs (...)
Mechanism (through which the target should be realised)	1	Quota
	1	Quota/Cost sharing/Information
Expected Impact	1	Reduction of municipal wastes
	2	Article 8 (1.) (...) strengthen the re-use and the prevention, recycling and other recovery of waste (...)
Year specified	1	2025/2035

	2	80 % when implemented after 4. July 2018 50% when implemented before 4. July 2018
Part of the value chain addressed	1	End of life
	2	Production

Policy document name	COUNCIL DIRECTIVE 1999/31/EC of 26 April 1999 on the landfill of waste (28)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	Restrict landfilling of waste recyclable or suitable for energy recovery
Mechanism (through which the target should be realised)	1	Ban
Expected Impact	1	(...) is to ensure a progressive reduction of landfilling of waste, in particular of waste that is suitable for recycling or other recovery, and, by way of stringent operational and technical requirements on the waste and land fills, to provide for measures, procedures and guidance (...)
Year specified	1	2030
Part of the value chain addressed	1	End of life

Policy document name	A sustainable bioeconomy for Europe - Strengthening the connection between economy, society and the environment : updated bioeconomy strategy (1)	
Qualitative		

Quantitative		
Goal qualitative description	1	(...) are managing natural resources sustainably, reducing dependence on non-renewable, unsustainable resources, limiting and adapting to climate change and strengthening European competitiveness and creating jobs. (...) 1% increase of European wood-based products in the market share of the global construction, textile and plastics markets could
Mechanism (through which the target should be realised)	1	-
Expected Impact	1	generate a revenue for the European wood-based bioeconomy in the scale of EUR 10 to 60 billion. This would amount to 3% - 20% of the current total turnover of the EU forest industry. The additional industrial roundwood use of this 1% increase would be at least 83 million cubic metres, which would be 23% of the total industrial roundwood production in the EU in 2016 (355 million cubic metres, data FAOSTAT).
Year specified	1	2050
Part of the value chain addressed	1	Consumption

Policy document name	Sustainable Carbon Cycles (25)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	Carbon farming should contribute -42 Mio. t CO ₂ eq to -310 Mio. t LULUCF target
Mechanism (through which the target should be realised)	1	Amount
Expected Impact	1	(...) increase carbon sink

D4.1. Report presenting scenarios and related policy packages

Year specified	1	2030
Part of the value chain addressed	1	Production

Policy document name	Pathway to a Healthy Planet for All EU Action Plan: 'Towards Zero Pollution for Air, Water and Soil' (22)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	Air, water and soil pollution is reduced to levels no longer considered harmful to health and natural ecosystems and that respect the boundaries our planet can cope with, thus creating a toxic-free environment.
	2	Under EU law, Green Deal ambitions and in synergy with other initiatives, by 2030 the EU should reduce: (...) 6. significantly total waste generation and by 50% residual municipal waste.
Mechanism (through which the target should be realised)	1	Planetary boundaries
	2	Quota
Expected Impact	1	"Air, water and soil pollution is reduced to levels no longer considered harmful to health and natural ecosystems and that respect the boundaries our planet can cope with"
	2	Reaching zero pollution targets by 2050
Year specified	1	2050
	2	2030
Part of the value chain addressed	1	End of life
	2	End of life

EU Chemical sector

Policy document name	Sustainable Carbon Cycles (25)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	By 2028, any ton of CO ₂ captured, transported, used and stored by industries should be reported and accounted by its fossil, biogenic or atmospheric origin;
	2	At least 20% of the carbon used in the chemical and plastic products should be from sustainable non-fossil sources by 2030, in full consideration of the EU's biodiversity and circular economy objectives and of the upcoming policy framework for bio-based, biodegradable and compostable plastics.
	3	5 Mt of CO ₂ should be annually removed from the atmosphere and permanently stored through frontrunner projects by 2030
Mechanism (through which the target should be realised)	1	Reporting
	2	Quota
	3	Quota
Expected Impact	1	(...)reduce our reliance on carbon
	2	(...) recycle carbon from waste streams, from sustainable sources of biomass or directly from the atmosphere, to use it in place of fossil carbon
	3	(...) we need to upscale carbon removal solutions that capture CO ₂ from the atmosphere and store it for the long term
Year specified	1	2028
	2	2030
	3	2030
	1	Production

Part of the value chain addressed	2	Production
	3	Production

Policy document name	Chemicals Strategy for Sustainability Towards a Toxic-Free Environment (30)	
Qualitative	X	
Quantitative	X	
Goal qualitative description	1	In a clean circular economy it is essential to boost the production and uptake of secondary raw materials and ensure that both primary and secondary materials and products are always safe. (...)However, the creation of a well-functioning market for secondary raw materials and the transition to safer materials and products is being slowed down by a number of issues, in particular the lack of adequate information on the chemical content of products (...).To move towards toxic-free material cycles and clean recycling and ensure that “Recycled in the EU” becomes a benchmark worldwide , it is necessary to ensure that substances of concern in products and recycled materials are minimised.(...)
	2	0% pollution for air, water, and soil
Mechanism (through which the target should be realised)	1	Ban, Assessment
	2	Quota
Expected Impact	1	(...) This includes ensuring that all chemicals are used more safely and sustainably, promoting that chemicals having a chronic effect for human health and the environment - substances of concern - are minimised and substituted as far as possible, and phasing out the most harmful ones for non-essential societal use, in particular in consumer products.(...)
	2	(...) aims to better protect citizens and the environment, and boost innovation for safe and sustainable chemicals, including in construction where chemicals are omnipresent.
Year specified	1	-
	2	2050

Part of the value chain addressed	1	Production
	2	Consumption

EU Construction sector

Policy document name	A renovation Wave for Europe greening our buildings, creating jobs, improving lives (35)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	2020: Annual renovation rate for energy renovation about 1% and deep renovation rate cutting 60 % of energy needs is about 0,2 % (Base: Building Stock 85% = 220 Mio. building units) 2030: At least 2% of annual energy renovation (160,000 jobs could be created, foster deep renovation rate); maximum about 35 Mio
Mechanism (through which the target should be realised)	1	(...) Communication presents a strategy to trigger a Renovation Wave for Europe, breaking down long-standing barriers to energy and resource-efficient renovation supporting fresh investment over a sustained period starting from public and less efficient buildings, spurring digitalisation and creating employment and growth opportunities across the renovation supply chain
Expected Impact	1	Increase in jobs and material usage
Year specified	1	2030
Part of the value chain addressed	1	Consumption

Policy document name	New EU Forest Strategy for 2030 (26)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	(...) Increase of market share for wood products for the support of reaching climate goals.

Mechanism (through which the target should be realised)	1	The Commission will develop a 2050 roadmap for reducing whole life-cycle carbon emissions in buildings. In the context of the revision of the Construction Products Regulation, the Commission will develop a standard, robust and transparent methodology to quantify the climate benefits of wood construction products and other building materials.
Expected Impact	1	(...) role of wood products is to help turn the construction sector from a source of greenhouse gas emissions into a carbon sink.
Year specified	1	2030
Part of the value chain addressed	1	Production

Policy document name	Proposal for a DIRECTIVE OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL on energy efficiency (recast) (20)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	(...) double their annual energy savings obligations, leading the way by action throughout the public sector. (...) renovate 3% of their buildings' floor space annually.
Mechanism (through which the target should be realised)	1	Quota
Expected Impact	1	The embodied carbon in construction is estimated to account for about 10% of total yearly greenhouse gas emissions worldwide, see IRP, Resource Efficiency and Climate Change, 2020, and UN Environment Emissions Gap Report 2019 (67 p. 37) Increase in material usage
Year specified	1	2030
Part of the value chain addressed	1	Production

Policy document name	Directive 2010/31/EE of the European Parliament and of the Council of 19 May 2010 on the energy performance of buildings (recast) (40)	
Qualitative	X	
Quantitative	X	
Goal qualitative description	1	Each Member State shall establish a long-term renovation strategy to support the renovation of the national stock of residential and non-residential buildings, both public and private, into a highly energy efficient and decarbonised building stock by 2050, facilitating the cost-effective transformation of existing buildings into nearly zero-energy buildings.
Mechanism (through which the target should be realised)	1	(...) Set out a roadmap with measures and domestically established measurable progress indicators, with a view to the long-term 2050 (...) The roadmap shall include indicative milestones for 2030, 2040 and 2050, and specify how they contribute to achieving the Union's energy efficiency targets in accordance with Directive 2012/27/EU
Expected Impact	1	(...) reducing greenhouse gas emissions in the Union by 80-95 % compared to 1990, in order to ensure a highly energy efficient and decarbonised national building stock and in order to facilitate the cost-effective transformation of existing buildings into nearly zero-energy buildings
Year specified	1	2050
Part of the value chain addressed	1	Production

Policy document name	Directive 2008/98/EC of the European Parliament and of the Council of 19 November 2008 on waste and repealing certain Directives (23)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	Article 11 (2a) "(...) (b) by 2020, the preparing for re-use, recycling and other material recovery, including backfilling operations using waste to substitute other materials, of non-hazardous construction and demolition waste excluding naturally occurring material defined in category 17 05 04 in the list of waste shall be increased to a minimum of 70 % by weight;"

Mechanism (through which the target should be realised)	1	Quota
Expected Impact	1	Article 11 (1.) (...) promote preparing for re-use activities, notably by encouraging the establishment of and support for preparing for re-use and repair networks, by facilitating, where compatible with proper waste management, their access to waste held by collection schemes or facilities that can be prepared for re-use but is not destined for preparing for re-use by those schemes or facilities, and by promoting the use of economic instruments, procurement criteria, quantitative objectives or other measures.(...)
Year specified	1	2020
Part of the value chain addressed	1	End of life

Policy document name	ANNEX to the COMMISSION DELEGATED REGULATION (EU) .../... supplementing Regulation (EU) 2020/852 of the European Parliament and of the Council by establishing the technical screening criteria for determining the conditions under which an economic activity qualifies as contributing substantially to the sustainable use and protection of water and marine resources, to the transition to a circular economy, to pollution prevention and control or to the protection and restoration of biodiversity and ecosystems and for determining whether that economic activity causes no significant harm to any of the other environmental objectives and amending Delegated Regulation (EU) 2021/2178 as regards specific public disclosures for those economic activities (38)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	4. The use of primary raw material in the construction of the building is minimised through the use of secondary raw materials. The operator of the activity ensures that the three heaviest material categories used to construct the building, measured by mass in kilogrammes, comply with the following maximum total amounts of primary raw material used: (a) for the combined total of concrete, natural or agglomerated stone, a maximum of 70% of the material come from primary raw material; (b) for the combined total of brick, tile, ceramic, a maximum of 70% of the material come from primary raw material; (c) for bio-based materials, a maximum of 80% of the total material come from primary raw material; (d) for the combined total of glass, mineral insulation, a maximum of 70% of the total material come from primary raw material;

		(e) for non-biobased plastic, a maximum of 50% of the total material come from primary raw material; (f) for metals, a maximum of 30% of the total material come from primary raw material; (g) for gypsum, a maximum of 65% of the material come from primary raw material.
	2	5. The use of primary raw material in the renovation of the building is minimised through the use of secondary raw materials. The operator of the activity ensures that the three heaviest material categories that have been newly added to the building in the renovation of the building, measured by mass in kilogrammes, comply with the following thresholds regarding the maximum amount of primary raw material used: (a) for the combined total of concrete, natural or agglomerated stone, a maximum of 85% of the material come from primary raw material; (b) for the combined total of brick, tile, ceramic, a maximum of 85% of the material come from primary raw material; (c) for bio-based materials, a maximum of 90% of the material come from primary raw material; (d) for the combined total of glass, mineral insulation, a maximum of 85% of the material come from primary raw material; (e) for non-biobased plastic, a maximum of 75% of the material come from primary raw material; (f) for metals, a maximum of 65% of the material come from primary raw material; (g) for gypsum, a maximum of 83% of the material come from primary raw material.
	3	4. The preparing for re-use or recycling of the non-hazardous construction and demolition waste generated on the construction site is at least 90% (by mass in kilogrammes), excluding backfilling. This excludes naturally occurring material referred to in category 17 05 04 in the European List of Waste established by Commission Decision 2000/532/EC. The operator of the activity demonstrates compliance with the 90% threshold by reporting (...). Alternatively, at least 95% of the mineral fraction and 70% of the non-mineral fraction of the non-hazardous demolition waste is separately collected and prepared for reuse or recycled.
Mechanism (through which the target should be realised)	1	Quota
	2	Quota

	3	Quota
Expected Impact	1	Substantial contribution to the transition to a circular economy
	2	Substantial contribution to the transition to a circular economy
	3	Substantial contribution to the transition to a circular economy
Year specified	1	-
	2	-
	3	-
Part of the value chain addressed	1	Production
	2	Production
	3	End of life

Policy document name	Expert knowledge without specific policy document reference based on the input of the INSPIRE conference	
Qualitative	X	
Quantitative		
Goal qualitative description	1	Oblige/stimulate insulation of all rented houses by 2030/2050
	2	Establishment of a (mandatory) Digital Product Passport for construction products on European level with - besides others - information regarding repair, reuse, recycling as well as biomass content.
	3	Buildings renovation quota/targets
	4	Incentives for the deployment of bio-based materials in construction

	5	Vacancy management to maintain old buildings and availability of buildings for prolonged usage
Mechanism (through which the target should be realised)	1	Regulatory
	2	Regulatory
	3	Quota
	4	Incentives
	5	Incentives
Expected Impact	1	Higher energy efficiency
	2	Consumer information
	3	Increase in energy efficiency
	4	Less fossil resource increase e in bio-based materials
	5	Less vacancies and less primary raw material usage
Year specified	1	2030
	2	2030
	3	2050
	4	2050
	5	2050
Part of the value chain addressed	1	Production
	2	Consumption

	3	Production
	4	Production
	5	Production

EU Plastic sector

Policy document name	Proposal for a REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL on packaging and packaging waste, amending Regulation (EU) 2019/1020 and Directive (EU) 2019/904, and repealing Directive 94/62/EC (44)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	Article 5 (5) (...) lower the sum of concentration levels of lead, cadmium, mercury and hexavalent chromium resulting from substances present in packaging or packaging components (...)
	2	Article 6 (1) All packaging shall be recyclable. Article 6 (3) Recyclable packaging shall, from 1 January 2030, comply with the design for recycling criteria as laid down in the delegated acts adopted pursuant to paragraph 4 and, from 1 January 2035, also with the recyclability at scale requirements laid down in the delegated acts adopted pursuant to paragraph 6.(...)
	3	Article 7 (1) From 1 January 2030, the plastic part in packaging shall contain the following minimum percentage of recycled content recovered from post-consumer plastic waste, per unit of packaging: (a) 30 % for contact sensitive packaging made from polyethylene terephthalate (PET) as the major component; (b) 10 % for contact sensitive packaging made from plastic materials other than PET, except single use plastic beverage bottles; (c) 30 % for single use plastic beverage bottles; (d) 35 % for packaging other than those referred to in points (a), (b) and (c)

		2. From 1 January 2040, the plastic part in packaging shall contain the following minimum percentage of recycled content recovered from post-consumer plastic waste, per unit of packaging: (a) 50 % for contact sensitive plastic packaging, except single use plastic beverage bottles; (b) 65 % for single use plastic beverage bottles; (c) 65 % for plastic packaging other than those referred to in points (a) and (b);
	4	Article 38 (1) Each Member State shall reduce the packaging waste generated per capita, as compared to the packaging waste generated per capita in 2018 as reported to the Commission in accordance with Decision 2005/270/EC, by (a) 5 % by 2030; (b) 10 % by 2035; (c) 15 % by 2040
	5	Article 46 (1) Member States shall take the necessary measures to attain the following recycling targets covering the whole of their territory: (a) by 31 December 2025, a minimum of 65 % by weight of all packaging waste generated; (b) by 31 December 2025, the following minimum percentages by weight of the following specific materials contained in packaging waste generated: (i) 50 % of plastic; (ii) 25 % of wood; (iii) 70 % of ferrous metals; (iv) 50 % of aluminium; (v) 70 % of glass; (vi) 75 % of paper and cardboard; (c) by 31 December 2030, a minimum of 70 % by weight of all packaging waste generated; (d) by 31 December 2030, the following minimum percentages by weight of the following specific materials contained in packaging waste generated: (i) 55 % of plastic; (ii) 30 % of wood; (iii) 80 % of ferrous metals; (iv) 60 % of aluminium; (v) 75 % of glass;

		(vi) 85 % of paper and cardboard
Mechanism (through which the target should be realised)	1	Quota/Reference levels
	2	Product specifications
	3	Quota
	4	Quota
	5	Quota
Expected Impact	All	(...) establishes requirements for the entire life cycle of packaging as regards environmental sustainability and labelling, to allow its placing on the market, as well as for the extended producer responsibility, collection, treatment and recycling of packaging waste (...) (...) contributes to the efficient functioning of the internal market by harmonising national measures on packaging and packaging waste in order to avoid obstacles to trade, distortion and restriction of competition within the Union (...)
	1	Article 5 (4) (...) They shall address, as appropriate, substances of concern that negatively affect the re-use and recycling of materials in the packaging in which they are present.(...)
	2	Higher percentage of recyclable packaging products
	3	Increase of raw material efficiency
	4	Waste prevention
	5	Recycling targets and promotion of recycling
Year specified	1	-
	2	2030/2035
	3	2030/2040
	4	2030/2035/2040

	5	2025/2030
Part of the value chain addressed	1	End of life
	2	Production
	3	Production
	4	End of life
	5	End of life

Policy document name	Directive 2008/98/EC of the European Parliament and of the Council of 19 November 2008 on waste and repealing certain Directives (23)	
Qualitative	X	
Quantitative		
Goal qualitative description	1	"Article 11 (1) " (...) member states shall set up separate collection at least for paper, metal, plastic and glass, and, by 1 January 2025, for textiles""
	2	Article 8a (1.) (...) Member States shall take the necessary measures to ensure that the financial contributions paid by the producer of the product to comply with its extended producer responsibility obligations: (a) cover the following costs for the products that the producer puts on the market in the Member State concerned: – costs of separate collection of waste and its subsequent transport and treatment, including treatment necessary to meet the Union waste management targets, and costs necessary to meet other targets and objectives as referred to in point (b) of paragraph 1, taking into account the revenues from re-use, from sales of secondary raw material from its products and from unclaimed deposit fees, – costs of providing adequate information to waste holders in accordance with paragraph 2, – costs of data gathering and reporting in accordance with point (c) of paragraph 1.
	1	Separation

Mechanism (through which the target should be realised)	2	Cost sharing
Expected Impact	1	“Article 11 (1.) Member States shall take measures to promote preparing for re-use activities, notably by encouraging the establishment of and support for preparing for re-use and repair networks, by facilitating, where compatible with proper waste management, their access to waste held by collection schemes or facilities that can be prepared for re-use but is not destined for preparing for re-use by those schemes or facilities, and by promoting the use of economic instruments, procurement criteria, quantitative objectives or other measures.
	2	“Article 8 (1.) In order to strengthen the re-use and the prevention, recycling and other recovery of waste, Member States may take legislative or non- legislative measures to ensure that any natural or legal person who professionally develops, manufactures, processes, treats, sells or imports products (producer of the product) has extended producer responsibility”
Year specified	1	2025
	2	-
Part of the value chain addressed	1	End of life
	2	End of life

Policy document name	ANNEX to the COMMISSION DELEGATED REGULATION (EU) .../... supplementing Regulation (EU) 2020/852 of the European Parliament and of the Council by establishing the technical screening criteria for determining the conditions under which an economic activity qualifies as contributing substantially to the sustainable use and protection of water and marine resources, to the transition to a circular economy, to pollution prevention and control or to the protection and restoration of biodiversity and ecosystems and for determining whether that economic activity causes no significant harm to any of the other environmental objectives and amending Delegated Regulation (EU) 2021/2178 as regards specific public disclosures for those economic activities (38)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	1. The activity complies with one of the following criteria: (a) use of circular feedstock: until 2028, at least 35% of the packaging product by weight consists of recycled post-consumer material for non-contact sensitive packaging and at least 10% for contact sensitive packaging.

		From 2028, at least 65% of the packaging product by weight consists of recycled post-consumer material for noncontact sensitive packaging and at least 50% for contact sensitive packaging; (b) design for reuse: the packaging product has been designed to be reusable within a reuse system and fulfils the requirements for the use of circular feedstock, as set in point 1.a with 35% and 10% targets for recycled feedstock applying as of 2028 and 65% and 50% targets applying as of 2032 (a) use of bio-waste feedstock: at least 65% of the packaging product by weight consists of sustainable bio-waste feedstock* (*Sustainable bio-waste feedstock refers to industrial bio-waste and municipal bio-waste, it excludes primary biomass in the absence of legally agreed sustainability criteria.)
Mechanism (through which the target should be realised)	1	Quota
Expected Impact	1	Substantial contribution to the transition to a circular economy
Year specified	1	2028/2032
Part of the value chain addressed	1	Production

Policy document name	Directive (EU) 2019/ of the European Parliament and of the Council of 5 June 2019 on the reduction of the impact of certain plastic products on the environment (43)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	Article 4 (1) Member States shall take the necessary measures to achieve an ambitious and sustained reduction in the consumption of the single-use plastic products listed in Part A of the Annex, in line with the overall objectives of the Union's waste policy, in particular waste prevention, leading to a substantial reversal of increasing consumption trends. Those measures shall achieve a measurable quantitative reduction in the consumption of the single-use plastic products listed in Part A of the Annex on the territory of the Member State by 2026 compared to 2022

	2	Article 5 Member States shall prohibit the placing on the market of the single-use plastic products listed in Part B of the Annex and of products made from oxo-degradable plastic.
	3	Article 6 (5) With regard to beverage bottles listed in Part F of the Annex, each Member State shall ensure that: (a) from 2025, beverage bottles listed in Part F of the Annex which are manufactured from polyethylene terephthalate as the major component ('PET bottles') contain at least 25 % recycled plastic, calculated as an average for all PET bottles placed on the market on the territory of that Member State; and (b) from 2030, beverage bottles listed in Part F of the Annex contain at least 30 % recycled plastic, calculated as an average for all such beverage bottles placed on the market on the territory of that Member State.
Mechanism (through which the target should be realised)	1	No specific
	2	Ban
	3	Quota
Expected Impact	1	Consumption reduction
	2	(...) to prevent and reduce the impact of certain plastic products on the environment, in particular the aquatic environment, and on human health
	3	(...) to promote the transition to a circular economy with innovative and sustainable business models, products and materials, thus also contributing to the efficient functioning of the internal market
Year specified	1	2026
	2	In force
	3	2025/2030
Part of the value chain addressed	1	Consumption
	2	Consumption
	3	Production

Policy document name	Expert knowledge without specific policy document reference based on the input of the INSPIRE conference	
Qualitative		
Quantitative	X	
Goal qualitative description	1	Less use of plastics in agriculture (mulch films, foil tunnels ect.)
	2	No fossil based plastic products till 2050 (recycled plastic and exchange through bio-based products, + CCU)
Mechanism (through which the target should be realised)	1	Ban
	2	Ban
Expected Impact	1	Reduction of negative impacts of fossil fuel use
	2	Reduction of negative impacts of fossil fuel use
Year specified	1	2030
	2	2050
Part of the value chain addressed	1	End of life
	2	End of life

EU Textile sector

Policy document name	EU Strategy for Sustainable and Circular Textiles (48)	
Qualitative	X	
Quantitative		

Goal qualitative description	1	"establishes the framework for setting ecodesign requirements for specific product categories in order to improve their circularity, energy performance and other environmental sustainability aspects."
	2	"(...) to support its uptake among producers and to offer consumers an easily recognisable and reliable way to choose eco-friendly products"
Mechanism (through which the target should be realised)	1	Certification
	2	Information
Expected Impact	1	"framework will enable the introduction of product performance and information requirements to be set for almost all categories of physical goods placed on the EU market."
	2	"to support its uptake among producers (...) "
Year specified	1	-
	2	-
Part of the value chain addressed	1	Production
	2	Consumption

Policy document name	Expert knowledge without specific policy document reference based on the input of the INSPIRE conference	
Qualitative	X	
Quantitative		
Goal qualitative description	1	Fortify local/ regional infrastructures to enable sharing of textiles (or other resources)
	2	Obligation for full recycling/reuse/refurbish of unsold textile/fashion (now 30% of fashion in never sold)

	3	supporting regenerative agricultural methods to grow/produce bio-based fibres (wool can be also used as long-term fertilizer)
	4	Increase incentives for bio-based materials in the sector
	5	Apply carbon tax on all fashion products (and other products) for GHG emissions in full product delivery chain
Mechanism (through which the target should be realised)	1	Establishing (markets, free sharing spaces, online platforms)
	2	Regulation
	3	Incentives
	4	Incentives
	5	Quota/Tax
Expected Impact	1	Reduction of reuse rather than disposal
	2	Reduction of waste and primary raw material usage
	3	Certified biobased materials/less land use impacts
	4	Higher percentage of bio-based material in the sector
	5	Reduction of GHG emission from production
Year specified	1	2025
	2	2030
	3	2030
	4	2050
	5	-

Part of the value chain addressed	1	End of life
	2	End of life
	3	Production
	4	Production
	5	Production

A3.2 Germany

Germany overarching goals

Policy document name	German Resource Efficiency Programme III - 2020 to 2023 (37)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	Further reduction of raw material consumption: 12 t as goal (2018 16,8 t) *aligned to https://www.umweltbundesamt.de/daten/umweltindikatoren/indikator-rohstoffkonsum#die-wichtigsten-fakten) Raise total raw material productivity: 1,5%/a “The goal of greater resource efficiency and resource substitution is to embed the principle of the circular or closed-loop economy in production processes, thus tapping into potentials for emission reduction that have not previously been used.” (NDC referring to resource efficiency program)
Mechanism (through which the target should be realised)	1	Raise raw material efficiency
Expected Impact	1	Reduction of primary resource usage
Year specified	1	-

Part of the value chain addressed	1	Production
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Policy document name	Act to Promote Circular Economy and Safeguard the Environmentally-Compatible Management of Waste (39)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	(1) The preparation for the re-use and recycling of municipal waste is to be: 1. at least 50 % by weight overall from 1 January 2020 onwards, 2. at least 55 % by weight overall from 1 January 2025 onwards, 3. at least 60 % by weight overall from 1 January 2030 onwards, and 4. at least 65 % by weight overall from 1 January 2035 onwards
Mechanism (through which the target should be realised)	1	Quota
Expected Impact	1	Higher re-use and recycling rate
Year specified	1	2025/2030/2035
Part of the value chain addressed	1	End of life

Germany construction sector

Policy document name	Integrated National Energy and Climate Plan (36)	
Qualitative	X	
Quantitative		
Goal qualitative description	1	Further savings could be achieved across various sectors by reducing indirect emissions, which occur during the production of building materials, components and plant technology, etc. in the industrial sector. In addition to promoting the use of

		resource-efficient building materials, the selective dismantling of buildings and the recycling of building materials could also contribute to reducing energy demand here. The Federal Government will submit an overall strategy for existing building stock in the form of the long-term renovation strategy pursuant to Article 2a of the Energy Performance of Buildings Directive. - increased funding for research and development projects, specialist and consumer information, competitions of ideas, model / demonstration projects aiming at - climate-friendly and innovative timber usage, especially in the area of building with wood, and in relation to the re-use of hardwood timber, the circular economy and cascading use - promoting climate-conscious consumer behaviour.
Mechanism (through which the target should be realised)	1	The Federal Government will submit an overall strategy for existing building stock in the form of the long-term renovation strategy pursuant to Article 2a of the Energy Performance of Buildings Directive. - increased funding for research and development projects, specialist and consumer information, competitions of ideas, model / demonstration projects aiming at - climate-friendly and innovative timber usage, especially in the area of building with wood, and in relation to the re-use of hardwood timber, the circular economy and cascading use promoting climate-conscious consumer behaviour.
Expected Impact	1	(...) savings could be achieved across various sectors by reducing indirect emissions, which occur during the production of building materials, components and plant technology.
Year specified	1	-
Part of the value chain addressed	1	Production

Policy document name	Act to Promote Circular Economy and Safeguard the Environmentally-Compatible Management of Waste (39)	
Qualitative	X	
Quantitative		
Goal qualitative description	1	70% by weight from 2020 prepared for recycling or other material recovery

Mechanism (through which the target should be realised)	1	Quota
Expected Impact	1	Increase in separation of material flows
Year specified	1	2020
Part of the value chain addressed	1	End of life

Germany plastic sector

Policy document name	Verpackungsabfälle (Packaging waste) (42)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	2025: Increase recycling quota for plastics: 3,8%/a 2030: Increase recycling quota for plastics: 8,8%/a
Mechanism (through which the target should be realised)	1	Quota
Expected Impact	1	Reaching goals of the packaging directive EU
Year specified	1	2025/2030
Part of the value chain addressed	1	End of life

Germany textile sector

Policy document name	Act to Promote Circular Economy and Safeguard the Environmentally-Compatible Management of Waste (39)	
Qualitative	X	
Quantitative		
Goal qualitative description	1	Separated collection of textiles from 2025
Mechanism (through which the target should be realised)	1	Regulation
Expected Impact	1	Increase in separation of material flows
Year specified	1	2025
Part of the value chain addressed	1	End of life

A3.3 Italy

Italy overarching goals

Policy document name	National Strategy for Circular Economy (51, 61)	
Qualitative	X	
Quantitative		
Goal qualitative description	1	the reform of the EPR and Consortia systems to support the achievement of EU targets through the creation of a specific supervisory body, under the presidency of the Ministry of Ecological Transition, to monitor the functioning and the effectiveness of Consortia;
	2	the development of a favourable fiscal system for a circular economy;

	3	the strengthening of upstream circularity strategies such as eco-design and extending the lifespan of products and parts;
	4	the development and uptake of methodologies that quantify all environmental impacts over the lifecycle of a product and throughout the waste management system;
	5	skills development for integrating circular economy issues in school curricula and professional training
	6	the development of secondary markets for raw materials supported by a clear framework for the supply of secondary raw materials and by an improved waste traceability system;
Mechanism (through which the target should be realised)	1	Monitor the functioning and effectiveness of consortia evaluating EPR schemes
	2	Provide fiscal and administrative facilities
	3	(...) promoting eco-innovation as a tool for competitiveness and sustainability and identifying tools to develop eco-innovation opportunities within the circular economy. (...) develop taxation supportive of the transition to a circular economy circular economy, to be achieved both by phasing out subsidies that are harmful to the environment
	4	(...) Implementing the principle of Extended Producer Responsibility so that it takes take responsibility for the final fate of the product up to the end-of-life stage, as well as the "Polluter Pays" principle (with schemes of vacuum returnable, pay-per-use, pay-as-you-throw)
	5	(...) introduce incentives in favour of those who promote behaviours individuals aimed at waste reduction, including consumers. (...) designing new consumer education and training programs interdisciplinary to the figure of circular economy expert, with the parallel development of facilities and public-private agreements for the development entrepreneurship in this new sector
	6	(...) competitive in terms of availability, performance and cost by acting on the materials purchasing chain (Minimum Environmental Criteria for Green Procurement in Public Administration), and on the criteria for the cassation of waste status ("End of Waste"). Essential to the creation of market conditions is the issue of traceability of materials and waste.
Expected Impact	1	-
	2	-

	3	-
	4	-
	5	Skills developed
	6	(...) create conditions for a market for secondary raw materials so that they are competitive (...)
Year specified	All	2035
Part of the value chain addressed	1	Production
	2	Production
	3	End of life
	4	Production
	5	Production
	6	Production
Policy document name	Bioeconomy in Italy (50)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	Increase by 20% the current output of the Italian bioeconomy (about €250 billion/year) and employment level (about 1.7 million)
Mechanism (through which the target should be realised)	1	Improving the sustainable production and quality of products in each of the sectors and interconnecting and leveraging the sectors more efficiently i) More investments in R&I, spin offs/start-ups, education, training, and communication, ii) better coordination between regional, national and EU stakeholders/policies,

		iii) better engagement with the public iv) tailored market development actions.
Expected Impact	1	Increase in employment
Year specified	1	2030
Part of the value chain addressed	1	Production

Policy document name	National Strategy for Sustainable Development (62)	
Qualitative	X	
Quantitative		
Goal qualitative description	1	14.1 By 2025, prevent and significantly reduce all forms of marine pollution, in particular from land-based activities, including marine litter and nutrient pollution. 11.6 By 2030, reduce the negative environmental impact per capita of cities, paying particular attention to air quality and the management of municipal and other waste. 12.5 By 2030, significantly reduce waste generation through prevention, reduction, recycling and reuse.
	2	12.2 Achieve sustainable management and efficient use of natural resources by 2030.
Mechanism (through which the target should be realised)	1	Quota
	2	-
Expected Impact	1	Achieving the national contribution for accomplishing the SDGs
	2	Achieving the national contribution for accomplishing the SDGs
Year specified	1	2025/2030
	2	2030

Part of the value chain addressed	1	End of life
	2	Production

Policy document name	Plan for the ecological transition (52)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	(...) an annual saving of annual savings of 20 Ktoe of energy and 0.04 MtCO ₂ in the non-ETS sectors.
	2	(...) Long-Term Strategy to 2050, amounting to 60% of the cut emissions for the residential and service sector, it is necessary to maintain a rate of retrofitting of buildings of almost 2% per year (...)
Mechanism (through which the target should be realised)	1	Quota
	2	Quota
Expected Impact	1	Reduction of energy usage in the non-ETS sectors
	2	Retrofitted building stock and less emissions
Year specified	1	-
	2	2050
Part of the value chain addressed	1	Production
	2	Production

Italy construction sector

Policy document name	Plan for the ecological transition (52)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	(...) an annual saving of annual savings of 20 Ktoe of energy and 0.04 MtCO ₂ in the non-ETS sectors.
	2	(...) Long-Term Strategy to 2050, amounting to 60% of the cut emissions for the residential and service sector, it is necessary to maintain a rate of retrofitting of buildings of almost 2% per year (...)
Mechanism (through which the target should be realised)	1	Quota
	2	Quota
Expected Impact	1	Reduction of energy usage in the non-ETS sectors
	2	Retrofitted building stock and less emissions
Year specified	1	-
	2	2050
Part of the value chain addressed	1	Production
	2	Production

Italy plastic sector

Policy document name	Plastic tax (63)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	From implementation 0,45 €/kg for • Single-use plastic items (MACSI) • Semi-finished products including preforms (e.g. sheets, plugs, bottles, films) • Empty and full packages • Not only food and beverage
Mechanism (through which the target should be realised)	1	Tax
Expected Impact	1	regulate the import and use of single-use plastic products standardized
Year specified	1	2024
Part of the value chain addressed	1	End of life

Italy textile sector

Policy document name	National recovery and resilience plan (49)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	Directive 851/2018 provides that for textiles, EU Member States have to establish separate collection by 1 January 2025. Italy, with Legislative Decree No. 116/2020, in transposing the provisions of the Directive, brought forward the deadline to 1 January 2022.

	2	target of 100% recovery in the textile sector, through 'Textile Hubs'
Mechanism (through which the target should be realised)	1	Quota
	2	Quota
Expected Impact	1	Less waste disposal
	2	support the improvement of the separate collection network in textile sector
Year specified	1	2022
	2	2026
Part of the value chain addressed	1	End of life
	2	End of life

A3.4 Netherlands

Netherlands overarching goals

Policy document name	National Circular Economy Programme 2023 - 2030 (31)	
Qualitative	X	
Quantitative	X	
Goal qualitative description	1	(...) reducing the raw material footprint coupled with reduction of the environmental impact: (...) reduction proposal in the EU taxonomy of 50% in 2030 and 75% in 2050 should be elaborated as new measure
	2	Levies for the use of primary fossil raw materials: at national and European level we will explore the possibilities for taxing nonenergetic use of primary fossil raw materials

Mechanism (through which the target should be realised)	1	Quota
	2	Levies
Expected Impact	1	All this will be done in a responsible way to protect our environment and health. This way, we will bring our use of raw materials back to a level that the Earth can handle.
	2	in order to stimulate the market for secondary raw materials
Year specified	1	2030/2050
	2	2030
Part of the value chain addressed	1	Production
	2	Production

Policy document name	Draft update of the National Plan Energy and Climate (32)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	In 2030: • consumption of primary raw materials is reduced by 50 %; • greenhouse gas emissions from production process and waste sector have been reduced to 36 Mton CO2 equivalent; (...) • biofeedstocks are considered to be standard
	2	In 2050, industrial raw materials, products and processes are net climate neutral and 80 % circular
Mechanism (through which the target should be realised)	1	Quota
	2	Quota

Expected Impact	1	Promoting clean energy technologies, long-term deployment of low-carbon technologies and related carbon transport and storage infrastructure
	2	Promoting clean energy technologies, long-term deployment of low-carbon technologies and related carbon transport and storage infrastructure
Year specified	1	2030
	2	2050
Part of the value chain addressed	1	Production
	2	Production

Netherlands construction sector

Policy document name	Expert knowledge without specific policy document reference based on the input of the INSPIRE conference	
Qualitative		
Quantitative	X	
Goal qualitative description	1	Guiding goal: by 2030, the Netherlands will use 50% less primary abiotic raw materials (minerals, metals and fossils).
	2	As many sustainable renewable raw materials are used as possible; products and raw materials are reused; and hardly any waste exists.
Mechanism (through which the target should be realised)	1	Quota
	2	-
Expected Impact	1	Increase in raw material efficiency
	2	Reduction of fossil fuels and wastes
Year specified	1	2030
	2	2050

Part of the value chain addressed	1	Production
	2	Production

Netherlands plastic sector

Policy document name	National Circular Economy Programme 2023 - 2030 (31)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	In 2030, the use of fossil raw materials for plastics will have been halved (compared to 2016) (...) The release of microplastics into the environment will have been reduced by at least 30%.
	2	In 2050 fewer plastics will be used - where this leads to environmental gains. (...) plastics will no longer be made from fossil raw materials, but rather from recycled raw materials, supplemented with secondary and sustainable bio-based raw materials and - over time - from carbon-based raw materials. Microplastics which are intentionally added to products will no longer exist. Emissions of microplastics (secondary or otherwise) into the environment will have been reduced by at least 70%, moreover.
Mechanism (through which the target should be realised)	1	to efficiency improvements and the use of mechanical recyclate, chemical recyclate and plastics based on sustainable bio-based raw materials
	2	plastics will no longer be made from fossil raw materials
Expected Impact	1	The ambition to close the plastic loop in 2050 and no longer produce plastics from primary fossil raw materials is an important dot on the horizon.
	2	Reaching the goal of 2050
Year specified	1	2030
	2	2050

Part of the value chain addressed	1	Production
	2	Production

Policy document name	Draft update of the National Plan Energy and Climate (32)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	The aim is to increase the obligation to 25 % -30 % plastic recycle or bio-based plastic by 2030. (In anticipation of EU legislation, a national obligation for plastic producers to incentivise the uptake of recycled or bio-based plastics will be introduced as of 2027.)
Mechanism (through which the target should be realised)	1	Quota
Expected Impact	1	This obligation applies to all plastics produced in the Netherlands and for the Dutch market. Exports are therefore excluded
Year specified	1	2027
Part of the value chain addressed	1	Production

Policy document name	Transition Agenda - Circular Economy (46)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	Annual reduction in growth of plastic: 1,5 %/a
	2	Annual increase in discarded plastics: 1,0 %/a

	3	Discarded plastics via route without sorting which are incinerated: 80%
	4	Discarded plastics via route without sorting which are exported: 2%
	5	Discarded plastics via route without sorting: 10%
	6	Sorted plastics as input for mechanical recycling: 50%
	7	Efficiency of mechanical recycling: 85%
	8	Chemically-recycled plastics: 10%
	9	Efficiency of chemical recycling: 60%
	10	Produced bioplastics: 15%
Mechanism (through which the target should be realised)	1	Prevention
	2	Prevention
	3	-
	4	More supply and demand
	5	Prevention
	6	Better quality, more environmental efficiency
	7	Better quality, more environmental efficiency
	8	More supply and demand
	9	-
	10	More supply and demand
Expected Impact	1	Annual reduction in growth of plastic

	2	Less disposal due to increase in products with long(er) lifespan
	3	-
	4	High usage value, effective return systems, litter control, enforcement
	5	Unsorted waste streams will be significantly reduced
	6	Increased quality of sorted plastics leads to more recycling and less incineration
	7	Increased quality of sorted plastics
	8	Growth in chemical recycling
	9	-
	10	Growth in (sustainably) produced bioplastics
Year specified	All	2030
Part of the value chain addressed	1	Consumption
	2	End of life
	3	End of life
	4	End of life
	5	End of life
	6	End of life
	7	Production
	8	Production
	9	Production

	10	Production
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Netherlands textile sector

Policy document name	National Circular Economy Programme 2023 - 2030 (31)	
Qualitative		
Quantitative	X	
Goal qualitative description	1	2025: • The share of recycled (post-consumer) / sustainable materials in textile products will be 25%. • 30% of resources, materials and products placed on the Dutch textile market will be recycled after collection - if direct reuse is no longer possible. • 10% of the textile products placed on the market in the Netherlands will be reused in the Netherlands after collection.
	2	2030: • In all textile products placed on the market in the Netherlands, 50% sustainable materials will have been used. Of that 50% at least 30% will be recycle and 20% will be sustainable textiles. • 50% of resources, materials and products placed on the Dutch textile market will be recycled after collection - if direct reuse is no longer possible. • 15% of the textile products placed on the market in the Netherlands will be reused in the Netherlands after collection.
	3	2035: The aim is to halve the ecological footprint of the textile industry with regard to emissions, water usage, chemicals and microplastics
	4	2050: Fully circular economy
	5	2030 New goals • New textiles placed on the market will have a circular design • Textiles will be used twice as long by consumers compared to 2023 • 100% of the textiles where lifetime extension is not possible will be collected and recycled

Mechanism (through which the target should be realised)	1	Quota
	2	Quota
	3	Quota
	4	Quota
	5	Quota
Expected Impact	1	Raw material efficiency increase and more share of recyclate in the markets
	2	Raw material efficiency increase and more share of recyclate in the markets
	3	halve the ecological footprint of the textile industry
	4	Reduction of raw material consumption and sustainable textile industry
	5	55% less impact on climate change; 55% less indirect land use; 50% less impact on eutrophication and acidification
Year specified	1	2025
	2	2030
	3	2035
	4	2050
	5	2030
Part of the value chain addressed	1	Production/End of life
	2	Production/End of life
	3	Production
	4	Production

	5	Production/Consumption
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A4 Policy portfolio scenarios

Table A4 3: Overview of portfolio of selected policy goals for the BioTransition scenario of the overarching EU level addressing CBBE aspects

Year	Policy goal description						
2023	Annual energy savings obligations by MSs: 2024-2030: >1.5% (20)						
2025	Increasing municipal waste recycling: >55% by weight by 2025 (23)						
2030	(...) Increase of market share for wood products for the support of reaching climate goals (26)	At least 32.5% improvement in energy efficiency by 2030; Reducing primary (39%) and final (36%) energy consumption by 2030 (20)	Net reduction of GHG emissions by at least 55% (By 2030 vs. 1990 level) (225 million tonnes of CO ₂ equivalent) (24)	(...) Overall Union renewable energy target to 42,5 % in the energy mix (50)	Restrict landfilling of waste recyclable or suitable for energy recovery (51)	Carbon farming should contribute -42 Mio. t CO ₂ eq to -310 Mio. t LULUCF target (25)	Under EU law, Green Deal ambitions and in synergy with other initiatives, by 2030 the EU should reduce: (...) 6. significantly total waste generation and by 50% residual municipal waste. (22)
2035	Increasing municipal waste recycling: >60% by weight by 2025 (23)						
2050	No net emission of greenhouse gas emissions in EU by 2050 where economic growth is decoupled from resource use (21)	(...) are managing natural resources sustainably, reducing dependence on non-renewable, unsustainable resources, limiting and adapting to climate change and strengthening European competitiveness and creating jobs. (1)	(...) 1% increase of European wood-based products in the market share of the global construction, textile and plastics markets could (1)	Air, water and soil pollution is reduced to levels no longer considered harmful to health and natural ecosystems and that respect the boundaries our planet can cope with, thus creating a toxic-free environment (22)			



D4.1. Report presenting scenarios and related policy packages



Table A4 4: Overview of portfolio of selected and more ambitious policy goals for the BioRevolution scenario of the overarching EU level addressing CBBE aspects

Year	Policy goal description						
2023	Annual energy savings obligations by MSs: 2024-2030: >1.5% (20)						
2025	Increasing municipal waste recycling: >55% by weight by 2025 (23)						
2030	(...) Increase of market share for wood products for the support of reaching climate goals (26)	At least 32.5% improvement in energy efficiency by 2030; Reducing primary (39%) and final (36%) energy consumption by 2030 (20)	Net reduction of GHG emissions by at least 55% (By 2030 vs. 1990 level) (225 million tonnes of CO2 equivalent) (24)	(...) Overall Union renewable energy target to 42,5 % in the energy mix (50)	Restrict landfilling of waste recyclable or suitable for energy recovery (51)	Carbon farming should contribute -42 Mio. t CO2eq to -310 Mio. t LULUCF target (25)	Under EU law, Green Deal ambitions and in synergy with other initiatives, by 2030 the EU should reduce: (...) 6. significantly total waste generation and by 50% residual municipal waste. (22)
2035	Increasing municipal waste recycling: >60% by weight by 2025 (23)						
2040	<i>Adapted or added policy goal:</i> <i>(...) economic growth is decoupled from resource use (21)</i>	<i>Adapted or added policy goal:</i> <i>(...) are managing natural resources sustainably, reducing dependence on non-renewable, unsustainable resources, limiting and adapting to climate change and strengthening European competitiveness and creating jobs. (1)</i>					

D4.1. Report presenting scenarios and related policy packages



2050	No net emission of greenhouse gas emissions in EU by 2050 (21)	<i>Adapted or added policy goal:</i> (...) Increase of market share for wood products for the support of reaching climate goals (...) (26) <i>To about 1,5% of worldwide market in the sectors of construction, plastics and textiles (adapted (1))</i>	Air, water and soil pollution is reduced to levels no longer considered harmful to health and natural ecosystems and that respect the boundaries our planet can cope with, thus creating a toxic-free environment (22)				
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Chemical sector

Table A4 5: Overview of portfolio of selected policy goals for the BioTransition scenario for the chemical sector addressing CBBE aspects

Year	Policy goal description			
2028	By 2028, any ton of CO2 captured, transported, used and stored by industries should be reported and accounted by its fossil, biogenic or atmospheric origin; (25)			
2030	At least 20% of the carbon used in the chemical and plastic products should be from sustainable non-fossil sources by 2030, in full consideration of the EU's biodiversity and circular economy objectives and of the upcoming policy framework for bio-based, biodegradable and compostable plastics. (25)	5 Mt of CO2 should be annually removed from the atmosphere and permanently stored through frontrunner projects by 2030. (25)	(...) reducing the raw material footprint coupled with reduction of the environmental impact: (...) reduction proposal in the EU taxonomy of 50% in 2030 (29)	(...) consumption of primary raw materials is reduced by 50 % (30)
2050	To move towards toxic-free material cycles and clean recycling and ensure that "Recycled in the EU" becomes a benchmark worldwide, it is necessary to ensure that substances of concern in products and recycled materials are minimised.(...) (28)	0% pollution for air, water, and soil (28)	(...) reducing the raw material footprint coupled with reduction of the environmental impact: (...) and 75% in 2050 should be elaborated as new measure (29)	In 2050, industrial raw materials, products and processes are net climate neutral and 80 % circular (30)

D4.1. Report presenting scenarios and related policy packages



Table A4 6: Overview of portfolio of selected and more ambitious policy goals for the BioRevolution scenario for the chemical sector addressing CBBE aspects

Year	Policy goal description			
2028	By 2028, any ton of CO2 captured, transported, used and stored by industries should be reported and accounted by its fossil, biogenic or atmospheric origin; (25)			
2030	<i>Adapted or added policy goal:</i> <i>At least 30% of the carbon used in the chemical and plastic products should be from sustainable non-fossil sources by 2030, and of 40% in 2050.</i> (...) in full consideration of the EU's biodiversity and circular economy objectives and of the upcoming policy framework for bio-based, biodegradable and compostable plastics. (25)	5 Mt of CO2 should be annually removed from the atmosphere and permanently stored through frontrunner projects by 2030. (25)	(...) reducing the raw material footprint coupled with reduction of the environmental impact: (...) reduction proposal in the EU taxonomy of 50% in 2030 (29)	(...) consumption of primary raw materials is reduced by 50 % (30)
2040	<i>Adapted or added policy goal:</i> <i>To move towards toxic-free material cycles and clean recycling and ensure that "Recycled in the EU" becomes a benchmark worldwide, it is necessary to ensure that substances of concern in products and recycled materials are minimised.(...) (28)</i>			
2045	<i>0% pollution for air, water, and soil (28)</i>			
2050	(...) reducing the raw material footprint coupled with reduction of the environmental	In 2050, industrial raw materials, products and processes are net climate neutral and 80 % circular (30)		



D4.1. Report presenting scenarios and related policy packages



	impact: (...) and 75% in 2050 should be elaborated as new measure (29)			
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Construction sector

Table A4 7: Overview of portfolio of selected policy goals for the BioTransition scenario for the construction sector addressing CBBE aspects

Year	Policy goal description					
2023	Article 11 (2a) "(...) (b) by 2020, the preparing for re-use, recycling and other material recovery shall be increased to a minimum of 70 % by weight;" (23)	4. The use of primary raw material in the construction is minimised through the use of secondary raw materials. (...) maximum total amounts of primary raw material used: (c) for bio-based materials, a maximum of 80% (...) come from primary raw material; (e) for non-biobased plastic, a maximum of 50% of the total material come from primary raw material; (40)	5. The use of primary raw material in the renovation of the building is minimised through the use of secondary raw materials. (...) maximum amount of primary raw material used: material come (c) for bio-based materials, a maximum of 90% of the material come from primary raw material; (e) for non-biobased plastic, a maximum of 75% of the material come from primary raw material; (40)	4. The preparing for re-use or recycling of (...) construction and demolition waste generated on the construction site is at least 90%. (...) Alternatively, at least 95% of the mineral fraction and 70% of the non-mineral fraction of the non-hazardous demolition waste is separately collected and prepared for reuse or recycled. (40)	GER: 70% by weight from 2020 prepared for recycling or other material recovery (60)	GER: Raise total raw material productivity 1,5%/a (35)
2030	At least 2% of annual energy renovation (33)	(...) double their annual energy savings obligations, leading the way by action throughout the public sector. (...) renovate 3% of their buildings' floor space annually. (20)	(...) Increase of market share for wood products for the support of reaching climate goals (...) role of wood products is to help turn the construction sector from a source of greenhouse gas emissions into a carbon sink. (26)	GER: will submit an overall strategy for existing building stock in the form of the long-term renovation strategy • climate-friendly and innovative timber usage, especially in the area of building with wood, and in relation to the re-use of hardwood timber, the circular economy and cascading use (34)		
2050	Each Member State shall establish a long-term renovation strategy to support the renovation of the national stock of residential and non-residential buildings, decarbonised building stock by 2050 (...) (55)					

Table A4 8: Overview of portfolio of selected and more ambitious policy goals for the BioRevolution scenario for the chemical sector addressing CBBE aspects

Year	Policy goal description									
2023	Article 11 (2a) "(...) (b) by 2020, the preparing for re-use, recycling and other material recovery shall be increased to a minimum of 70 % by weight;" (23)	GER: 70% by weight from 2020 prepared for recycling or other material recovery (60)	Adapted or added policy goal GER: Raise total raw material productivity: 2%/a (35)							
2025	Adapted or added policy goal specific capacity buildings activities for the professionals and raise their awareness									

D4.1. Report presenting scenarios and related policy packages



2030	At least 2% of annual energy renovation (33) <i>Adapted or added policy goal</i> <i>That includes obligation for insulation of all rented houses</i>	(...) double their annual energy savings obligations, leading the way by action throughout the public sector. (...) renovate 3% of their buildings' floor space annually. (20)	(...) role of wood products is to help turn the construction sector from a source of greenhouse gas emissions into a carbon sink. (26)	GER: will submit an overall strategy for existing building stock in the form of the long-term renovation strategy • climate-friendly and innovative timber usage, especially in the area of building with wood, and in relation to the re-use of hardwood timber, the circular economy and cascading use (34)	<i>Adapted or added policy goal</i> <i>5 Mt of CO2 should be annually removed from the atmosphere and permanently stored through frontrunner projects by 2030. (25)</i> <i>(including vacancy management of buildings)</i>	<i>Adapted or added policy goal</i> <i>The use of primary raw material in the construction is minimised through the use of secondary raw materials. (...) maximum total amounts of primary raw material used: (c) for bio-based materials, a maximum of 80% (...) come from primary raw material; (e) for non-biobased plastic, a maximum of 50% of the total material come from primary raw material; (40)</i>	<i>Adapted or added policy goal</i> <i>The use of primary raw material in the renovation of the building is minimised through the use of secondary raw materials. (...) maximum amount of primary raw material used: material come (c) for bio-based materials, a maximum of 90% of the material come from primary raw material; (e) for non-biobased plastic, a maximum of 75% of the material come from primary raw material; (40)</i>	<i>Adapted or added policy goal</i> <i>The preparing for re-use or recycling of (...) construction and demolition waste generated on the construction site is at least 90%. (...) Alternatively, at least 95% of the mineral fraction and 70% of the non-mineral fraction of the non-hazardous demolition waste is separately collected and prepared for reuse or recycled. (40)</i>	<i>Adapted or added policy goal</i> <i>Establishment of a (mandatory) Digital Product Passport for construction products on European level with - besides others - information regarding repair, reuse, recycling as well as biomass content.</i>	<i>Adapted or added policy goal</i> <i>Guiding goal: by 2030, the Netherlands will use 50% less primary abiotic raw materials (minerals, metals and fossils).</i>
2050	Each Member State shall establish a long-term renovation strategy to support the renovation of the national stock of residential and non-residential buildings, decarbonised building stock by 2050 (...) (55)	<i>Adapted policy goal:</i> (...) Increase of market share for wood products for the support of reaching climate goals (...) (26) <i>To about 1,5% of worldwide market in the sectors of construction, plastics and textiles</i>	<i>Adapted policy goal:</i> <i>as many sustainable renewable raw materials are used as possible; products and raw materials are reused; and hardly any waste exists.</i>							



Plastic sector

Table A4 9: Overview of portfolio of selected policy goals for the BioTransition scenario for the plastic sector addressing CBBE aspects

Year	Policy goal description				
2023	Article 5 (5) (...) lower the sum of concentration levels of lead, cadmium, mercury and hexavalent chromium resulting from substances present in packaging or packaging components (...) (39)	"Article 11 (1) " (...) member states shall set up separate collection at least for paper, metal, plastic and glass, (...) (23)	Article 5 Member States shall prohibit the placing on the market of the single-use plastic products listed in Part B of the Annex and of products made from oxo-degradable plastic. (38)	1. The activity complies with one of the following criteria: (a) use of circular feedstock: until 2028, at least 35% of the packaging product by weight consists of recycled post-consumer material for non-contact sensitive packaging and at least 10% for contact sensitive packaging (40)	GER: Raise total raw material productivity 1,5%/a (35)
2025	Article 46 (1) Member States shall take the necessary measures to attain the following recycling targets (a) by 31 December 2025, a minimum of 65 % by weight of all packaging waste generated; (b) by 31 December 2025, the following minimum percentages by weight of the following specific materials contained in packaging waste generated: (i) 50 % of plastic; (ii) 25 % of wood; (39)	Article 6 (5) With regard to beverage bottles listed in Part F of the Annex: (a) from 2025, beverage bottles (...) manufactured from polyethylene terephthalate as the major component ('PET bottles') contain at least 25 % recycled plastic (...) (38)	Article 11 (1) " (...) member states shall set up separate collection (...) (23)	GER: Increase recycling quota for plastics: 3,8%/a (37)	
2028	1. The activity complies with one of the following criteria: From 2028, at least 65% of the packaging product by weight consists of recycled post-consumer material for noncontact sensitive packaging and at least 50% for contact sensitive packaging; (a) use of bio-waste feedstock: at least 65% of the packaging product by weight consists of sustainable bio-waste feedstock (40)	Article 6 (5) (...) (b) from 2030, beverage bottles listed in Part F of the Annex contain at least 30 % recycled plastic (...) (38)			

D4.1. Report presenting scenarios and related policy packages



2030	Article 6 (3) Recyclable packaging shall, from 1 January 2030, comply with the design for recycling criteria as laid down in the delegated acts adopted pursuant to paragraph 4 and, (39)	Article 7 (1) From 1 January 2030, the plastic part in packaging shall contain the following minimum percentage of recycled content recovered from post-consumer plastic waste, per unit of packaging: (a) 30 % for contact sensitive packaging made from polyethylene terephthalate (PET) as the major component; (b) 10 % for contact sensitive packaging made from plastic materials other than PET, except single use plastic beverage bottles; (c) 30 % for single use plastic beverage bottles; (d) 35 % for packaging other than those referred to in points (a), (b) and (c) (39)	Article 38 (1) Each Member State shall reduce the packaging waste generated per capita, by (a) 5 % by 2030; (b) 10 % by 2035; (c) 15 % by 2040 (39)	Article 46 (1) Member States shall take the necessary measures to attain the following recycling targets (c) by 31 December 2030, a minimum of 70 % by weight of all packaging waste generated; (d) by 31 December 2030, the following minimum percentages by weight of the following specific materials contained in packaging waste generated: (i) 55 % of plastic; (ii) 30 % of wood; (39)	GER: Increase recycling quota for plastics: 8,8%/a (37)
2035	Article 6 (3) from 1 January 2035, also with the recyclability at scale requirements (39)				
2040	2. From 1 January 2040, the plastic part in packaging shall contain the following minimum percentage of recycled content recovered from post-consumer plastic waste, per unit of packaging: (a) 50 % for contact sensitive plastic packaging, except single use plastic beverage bottles; (b) 65 % for single use plastic beverage bottles; (c) 65 % for plastic packaging other than those referred to in points (a) and (b); (39)				



D4.1. Report presenting scenarios and related policy packages



Table A4 10: Overview of portfolio of selected and more ambitious policy goals for the BioRevolution scenario for the plastic sector addressing CBBE aspects

Year	Policy goal description						
2023	Article 5 (5) (...) lower the sum of concentration levels of lead, cadmium, mercury and hexavalent chromium resulting from substances present in packaging or packaging components (...) (39)	"Article 11 (1) " (...) member states shall set up separate collection at least for paper, metal, plastic and glass, (...) (23)	Article 5 Member States shall prohibit the placing on the market of the single-use plastic products listed in Part B of the Annex and of products made from oxo-degradable plastic. (38)	1. The activity complies with one of the following criteria: (a) use of circular feedstock: until 2028, at least 35% of the packaging product by weight consists of recycled post-consumer material for non-contact sensitive packaging and at least 10% for contact sensitive packaging (40)	<i>Adapted or added policy goal:</i> <i>GER: Raise total raw material productivity: 2%/a (35)</i>		
2025	Article 46 (1) Member States shall take the necessary measures to attain the following recycling targets (a) by 31 December 2025, a minimum of 65 % by weight of all packaging waste generated; (b) by 31 December 2025, the following minimum percentages by weight of the following specific materials contained in packaging waste generated: (i) 50 % of plastic; (ii) 25 % of wood; (39)	Article 6 (5) With regard to beverage bottles listed in Part F of the Annex: (a) from 2025, beverage bottles (...) manufactured from polyethylene terephthalate as the major component ('PET bottles') contain at least 25 % recycled plastic (...) (38)	Article 11 (1) " (...) member states shall set up separate collection (...) (23)	GER: Increase recycling quota for plastics: 3,8%/a (37)	<i>Adapted or added policy goal:</i> <i>Article 6 (3) Recyclable packaging shall, from 1 January 2030, comply with the design for recycling criteria as laid down in the delegated acts adopted pursuant to paragraph 4 and, (39)</i>	<i>Adapted or added policy goal:</i> <i>Annual reduction in growth of plastic: 1,5 %/a (42)</i>	<i>Adapted or added policy goal:</i> <i>Article 6 (3) Recyclable packaging shall, from 1 January 2030, comply with the design for recycling criteria as laid down in the delegated acts adopted pursuant to paragraph 4 and, (39)</i>

D4.1. Report presenting scenarios and related policy packages



2028	<i>1. The activity complies with one of the following criteria: From 2028, at least 65% of the packaging product by weight consists of recycled post-consumer material for noncontact sensitive packaging and at least 50% for contact sensitive packaging; (a) use of bio-waste feedstock: at least 65% of the packaging product by weight consists of sustainable bio-waste feedstock (40)</i>	Article 6 (5) (...) (b) from 2030, beverage bottles listed in Part F of the Annex contain at least 30 % recycled plastic (...) (38)					
2030	Article 7 (1) From 1 January 2030, the plastic part in packaging shall contain the following minimum percentage of recycled content recovered from post-consumer plastic waste, per unit of packaging: (a) 30 % for contact sensitive packaging made from polyethylene terephthalate (PET) as the major component; (b) 10 % for contact sensitive packaging made from plastic materials other than PET, except single use plastic beverage bottles; (c) 30 % for single use plastic beverage bottles; (d) 35 % for packaging other than those referred to in points (a), (b) and (c) (39)	<i>Adapted or added policy goal:</i> Article 46 (1) Member States shall take the necessary measures to attain the following recycling targets (c) by 31 December 2030, a minimum of 70 % by weight of all packaging waste generated; (d) by 31 December 2030, the following minimum percentages by weight of the following specific materials contained in packaging waste generated: (39) <i>(i) 60 % of plastic; (ii) 35 % of wood;</i>	GER: Increase recycling quota for plastics: 8,8%/a (37)	<i>Adapted or added policy goal:</i> <i>Article 38 (1) Each Member State shall reduce the plastic waste generated per capita, by</i> <i>(a) 7,5 % by 2030; (b) 10 % by 2035; (c) 15 % by 2040 (39)</i>	<i>Adapted or added policy goal:</i> <i>less use of plastics in agriculture (mulch films, foil tunnels ect.)</i>		
2035	Article 6 (3) from 1 January 2035, also with the recyclability at scale requirements (39)						

D4.1. Report presenting scenarios and related policy packages



2040	2. From 1 January 2040, the plastic part in packaging shall contain the following minimum percentage of recycled content recovered from post-consumer plastic waste, per unit of packaging: (a) 50 % for contact sensitive plastic packaging, except single use plastic beverage bottles; (b) 65 % for single use plastic beverage bottles; (c) 65 % for plastic packaging other than those referred to in points (a) and (b); (39)	<i>Adapted or added policy goal:</i> <i>Mechanical recycling: 45% and chemical recycling of 15% (adapted of (27, 42))</i>					
2050	<i>Adapted or added policy goal:</i> <i>As much as possible no fossil based plastic products till 2050 (recycled plastic and exchange through bio-based products, + CCU)</i>						



D4.1. Report presenting scenarios and related policy packages



Textile sector

Table A4 11: Overview of portfolio of selected policy goals for the BioTransition scenario for the textile sector addressing CBBE aspects

Year	Policy objective description					
2023	ITA: Directive 851/2018 provides that for textiles, EU Member States have to establish separate collection by 1 January 2025. Italy, with Legislative Decree No. 116/2020, in transposing the provisions of the Directive, brought forward the deadline to 1 January 2022. (45)	ITA: (...) an annual saving of 20 Ktoe of energy and 0.04 MtCO ₂ in the non-ETS sectors. (48)				
2025	Article 11 (1) " (...) member states shall set up separate collection (...) and, by 1 January 2025, for textiles" (23)					
2026	Target of 100% recovery in the textile sector, through "Textile Hubs" (45)					
2030	"establishes the framework for setting eco-design requirements for specific product categories in order to improve their circularity, energy performance and other environmental sustainability aspects." (44)	" (...) to support its uptake among producers and to offer consumers an easily recognisable and reliable way to choose eco-friendly products" (44)	ITA: Increase by 20% the current output of the Italian bioeconomy (about €250 billion/year) and employment level (about 1.7 million) (46)			
2035	ITA: the reform of the EPR and Consortia systems to support the achievement of EU targets through the creation of a specific supervisory body, under the presidency of the Ministry of Ecological Transition, to monitor	ITA: the development of a favourable fiscal system for a circular economy;(47)	ITA: the strengthening of upstream circularity strategies such as eco-design and extending the lifespan of products and parts;(47)	ITA: the development and uptake of methodologies that quantify all environmental impacts over the lifecycle of a product and throughout the waste management system;(47)	ITA: skills development for integrating circular economy issues in school curricula and professional training(47)	ITA: the development of secondary markets for raw materials supported by a clear framework for the supply of secondary raw materials and by

D4.1. Report presenting scenarios and related policy packages



	the functioning and the effectiveness of Consortia;(47)					an improved waste traceability system;(47)
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Table A4 12: Overview of portfolio of selected and more ambitious policy goals for the BioRevolution scenario for the textile sector addressing CBBE aspects

Year	Policy objective description							
2023	ITA: Directive 851/2018 provides that for textiles, EU Member States have to establish separate collection by 1 January 2025. Italy, with Legislative Decree No. 116/2020, in transposing the provisions of the Directive, brought forward the deadline to 1 January 2022. (45)	ITA: (...) an annual saving of 20 Ktoe of energy and 0.04 MtCO ₂ in the non-ETS sectors. (48)						
2025	Article 11 (1) " (...) member states shall set up separate collection (...) and, by 1 January 2025, for textiles" (23)	<i>Adapted or added policy goal:</i> <i>Fortify local / regional infrastructures to enable sharing of textiles (or other resources) (markets, free sharing spaces, online platforms)</i>						
2026	Target of 100% recovery in the textile sector, through "Textile Hubs" (45)							
2027	<i>Adapted or added policy goal:</i> <i>The share of recycled (post-consumer) / sustainable materials in textile products will be 25%.</i>	<i>Adapted or added policy goal:</i> <i>30% of resources, materials and products placed on the textile market will be recycled after collection - if direct reuse is no longer possible.</i>	<i>Adapted or added policy goal:</i> <i>10% of the textile products placed on the market will be reused after collection.</i>					

D4.1. Report presenting scenarios and related policy packages



2030	"establishes the framework for setting eco-design requirements for specific product categories in order to improve their circularity, energy performance and other environmental sustainability aspects." (44)	"(...) to support its uptake among producers and to offer consumers an easily recognisable and reliable way to choose eco-friendly products" (44)	ITA: Increase by 20% the current output of the Italian bioeconomy (about €250 billion/year) and employment level (about 1.7 million) (46)	<i>Adapted or added policy goal:</i> <i>In all textile products placed on the market 50% sustainable materials will have been used. Of that 50% at least 30% will be recycled and 20% will be sustainable textiles.</i>	<i>Adapted or added policy goal:</i> <i>50% of resources, materials and products placed on the market will be recycled after collection - if direct reuse is no longer possible.</i>	<i>Adapted or added policy goal:</i> <i>15% of the textile products placed on the market will be reused in the after collection.</i>	<i>Adapted or added policy goal:</i> <i>Obligation for full recycling/reuse/refurbish of unsold textile/fashion</i>	<i>Adapted or added policy goal:</i> <i>supporting regenerative agricultural methods to grow/produce bio-based fibres (wool can be also used as long-term fertilizer)</i>
2035	ITA: the reform of the EPR and Consortia systems to support the achievement of EU targets through the creation of a specific supervisory body, under the presidency of the Ministry of Ecological Transition, to monitor the functioning and the effectiveness of Consortia;(47)	ITA: the development of a favourable fiscal system for a circular economy;(47)	ITA: the strengthening of upstream circularity strategies such as eco-design and extending the lifespan of products and parts;(47)	ITA: the development and uptake of methodologies that quantify all environmental impacts over the lifecycle of a product and throughout the waste management system;(47)	ITA: skills development for integrating circular economy issues in school curricula and professional training(47)	ITA: the development of secondary markets for raw materials supported by a clear framework for the supply of secondary raw materials and by an improved waste traceability system;(47)	<i>Adapted or added policy goal:</i> <i>The aim is to halve the ecological footprint of the textile industry with regard to emissions, water usage, chemicals and microplastics</i>	<i>Adapted or added policy goal:</i> <i>Apply carbon tax on all fashion products (and other products) for GHG emissions in full product delivery chain</i>
2050	<i>Adapted or added policy goal:</i> <i>Full circular textile material flows</i>	<i>Adapted or added policy goal</i> <i>Increase incentives for bio-based materials in the sector</i>						